

CIBMTR Reporting FHIR IG

8/5/21

FHIR Implementation Guide

An implementation guide (IG) is a set of rules about how FHIR resources are used (or should be used) to solve a particular problem, with associated documentation to support and clarify the usage. Classically, FHIR implementation guides are published on the web after they are generated using the FHIR Implementation Guide Publisher

used to inform

used to validate



FHIR Implementation Guides

- Are packages of FHIR conformance resources relevant to solving a particular interoperability need
- Also include additional human-readable documentation to guide implementation
- Provide support for computable validation of solutions that claim conformance with the IG
- Can build on other implementation guides
- Are defined by a FHIR ImplementationGuide resource



FHIR Implementation Guides

- Contain hundreds of files
  - xml, json, html, css, png, gif, zip, xlsx, ttl, sch, csv, php
  - representing
    - Narratives, Profiles, Extensions, Code Systems, Value Sets, Logical Models, Schemas, QA results, Examples, etc
  - versioning
- Generated by special tooling
  - Sushi
    - preprocessor
  - IG Publisher
    - creates final content from input files and templates

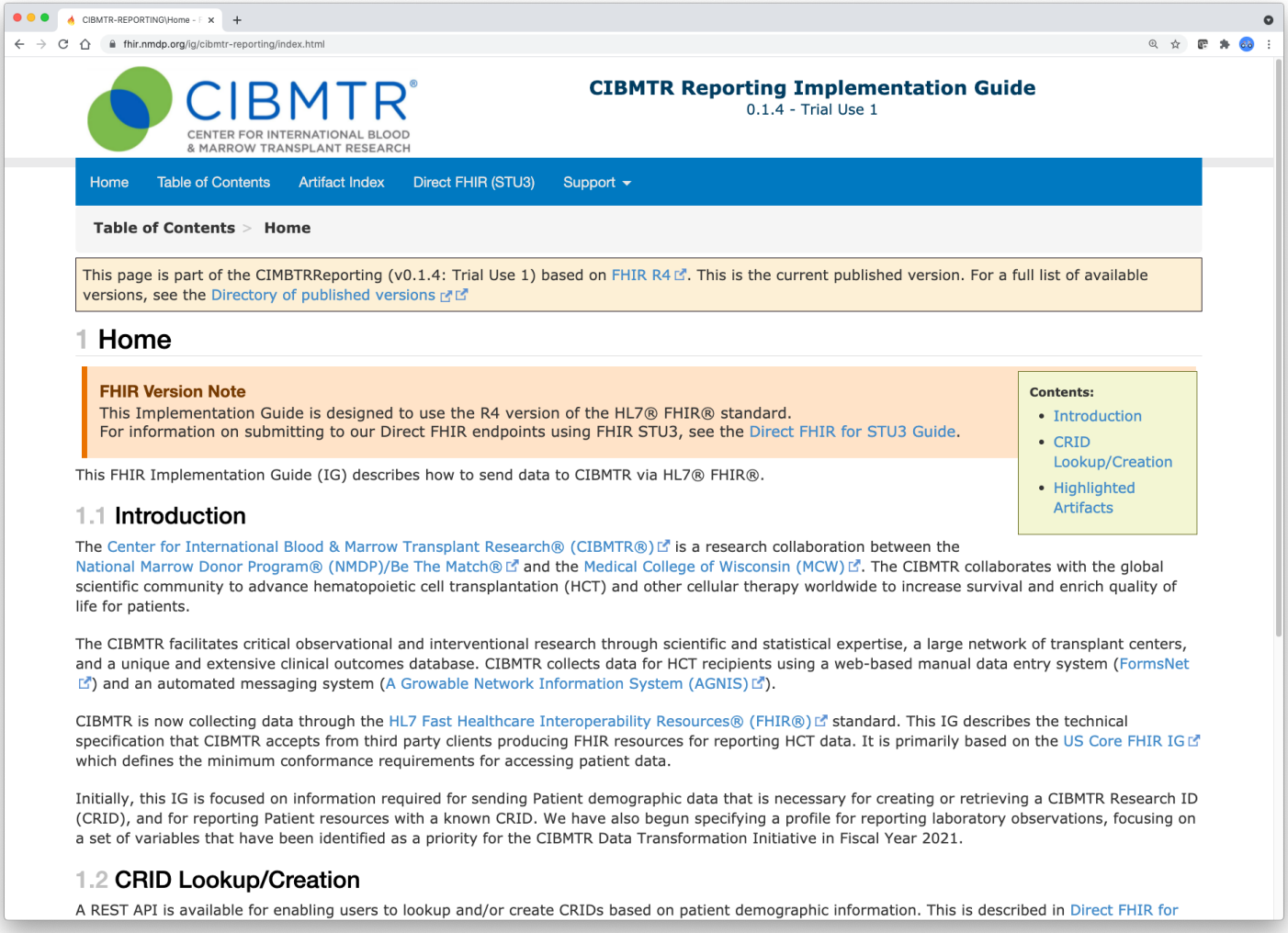


CIBMTR Reporting FHIR IG

http://fhir.nmdp.org/ig/cibmtr-reporting

- Based on HL7 FHIR R4
- Profiles
  - Patient (derived from US Core Patient)
  - Observation
    - Laboratory results (derived from US Core Lab Observation)
    - Priority Variables (derived from CIBMTR Laboratory Results)
- ValueSets
  - e.g., Priority Variables for FY 2021
- Examples
- Documentation
  - e.g., Direct FHIR Connection Guide





CIBMTR-REPORTING\Home -

fhir.nmdp.org/ig/cibmtr-reporting/index.html

CIBMTR®
CENTER FOR INTERNATIONAL BLOOD & MARROW TRANSPLANT RESEARCH

CIBMTR Reporting Implementation Guide
0.1.4 - Trial Use 1

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This page is part of the CIMBTRReporting (v0.1.4: Trial Use 1) based on FHIR R4. This is the current published version. For a full list of available versions, see the Directory of published versions

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FHIR Version Note
This Implementation Guide is designed to use the R4 version of the HL7® FHIR® standard.
For information on submitting to our Direct FHIR endpoints using FHIR STU3, see the Direct FHIR for STU3 Guide.

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- Introduction
- CRID Lookup/Creation
- Highlighted Artifacts

This FHIR Implementation Guide (IG) describes how to send data to CIBMTR via HL7® FHIR®.

1.1 Introduction

The Center for International Blood & Marrow Transplant Research® (CIBMTR®) is a research collaboration between the National Marrow Donor Program® (NMDP)/Be The Match® and the Medical College of Wisconsin (MCW). The CIBMTR collaborates with the global scientific community to advance hematopoietic cell transplantation (HCT) and other cellular therapy worldwide to increase survival and enrich quality of life for patients.

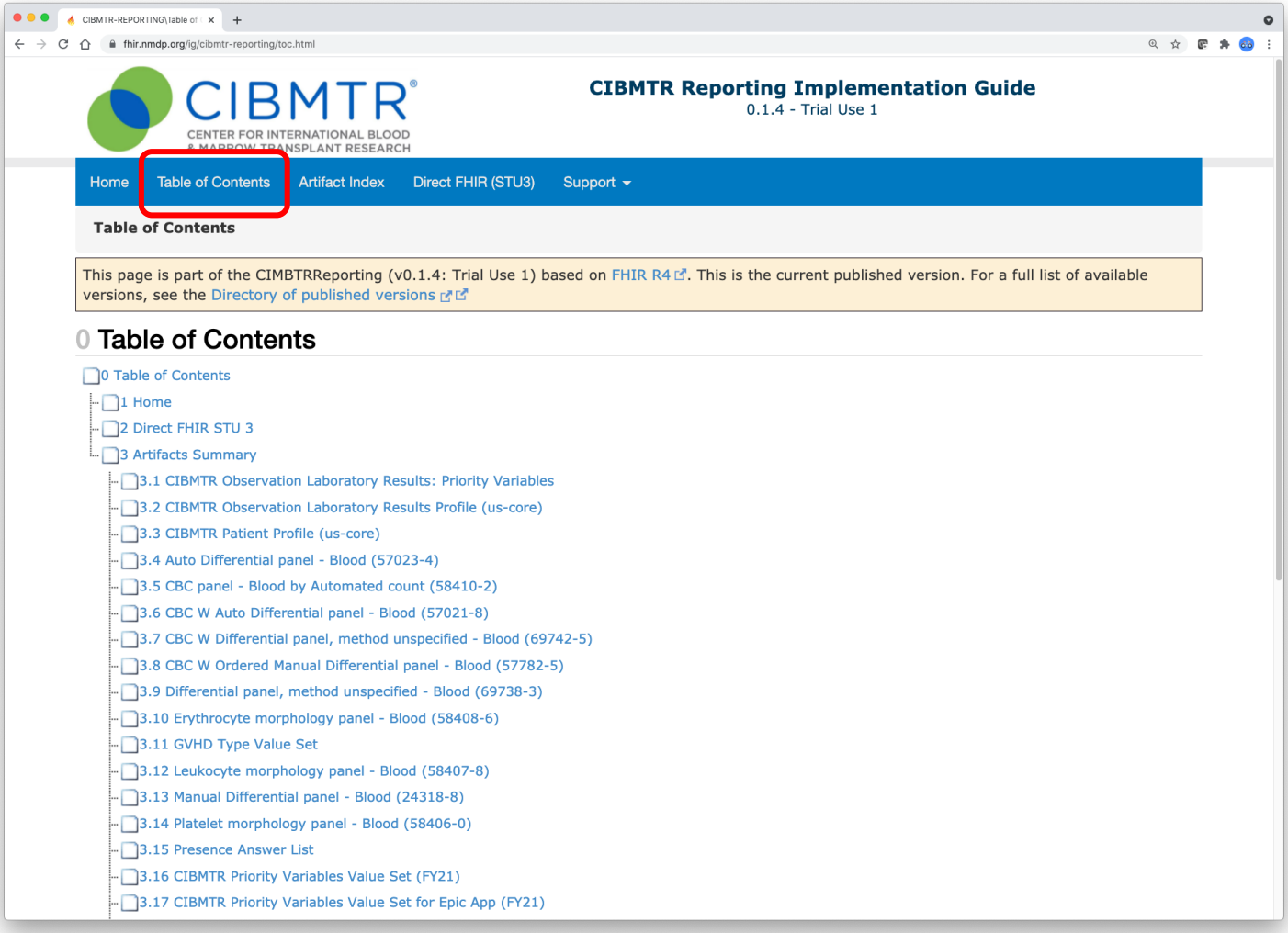
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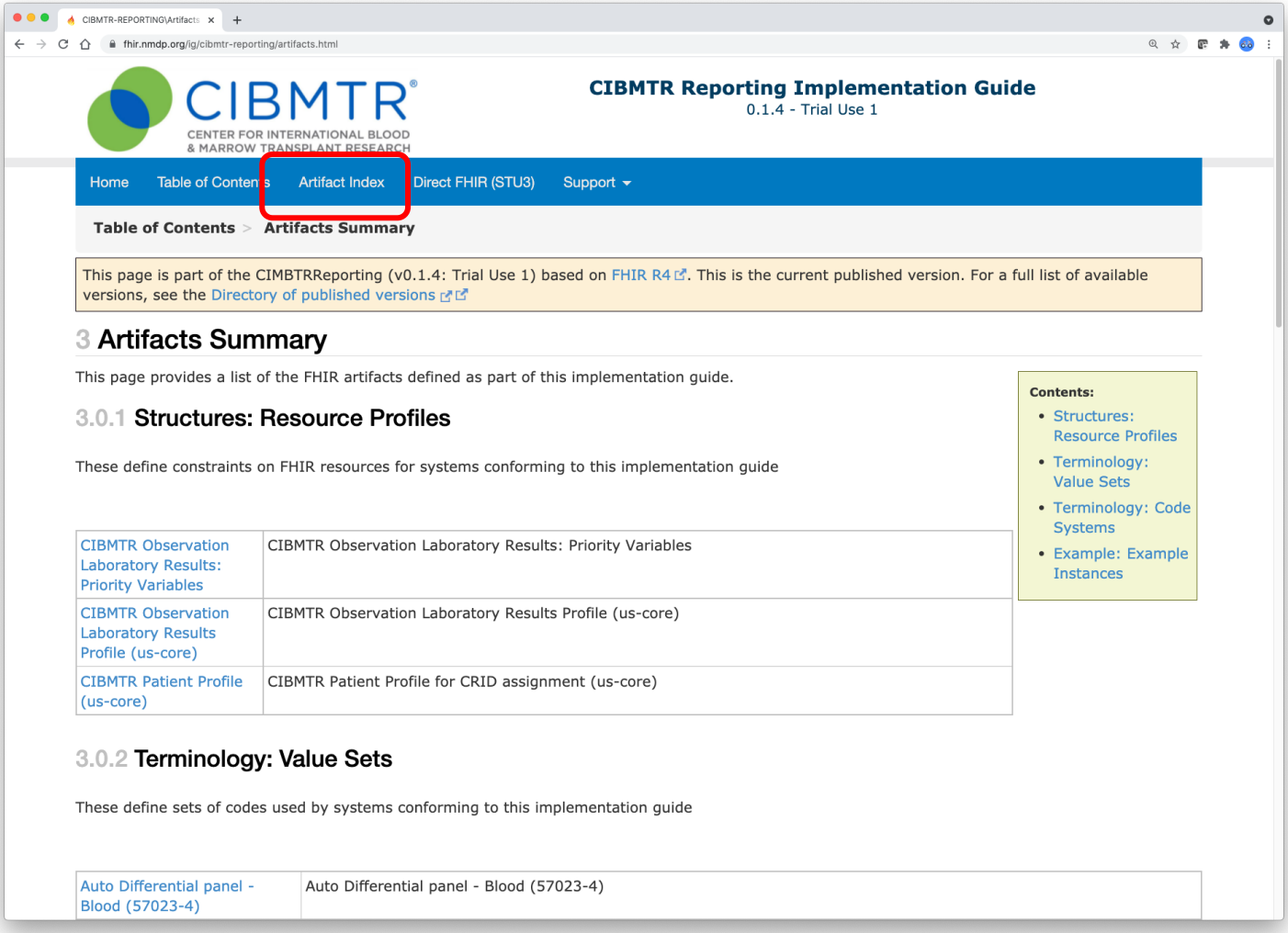
CIBMTR is now collecting data through the HL7 Fast Healthcare Interoperability Resources® (FHIR®) standard. This IG describes the technical specification that CIBMTR accepts from third party clients producing FHIR resources for reporting HCT data. It is primarily based on the US Core FHIR IG which defines the minimum conformance requirements for accessing patient data.

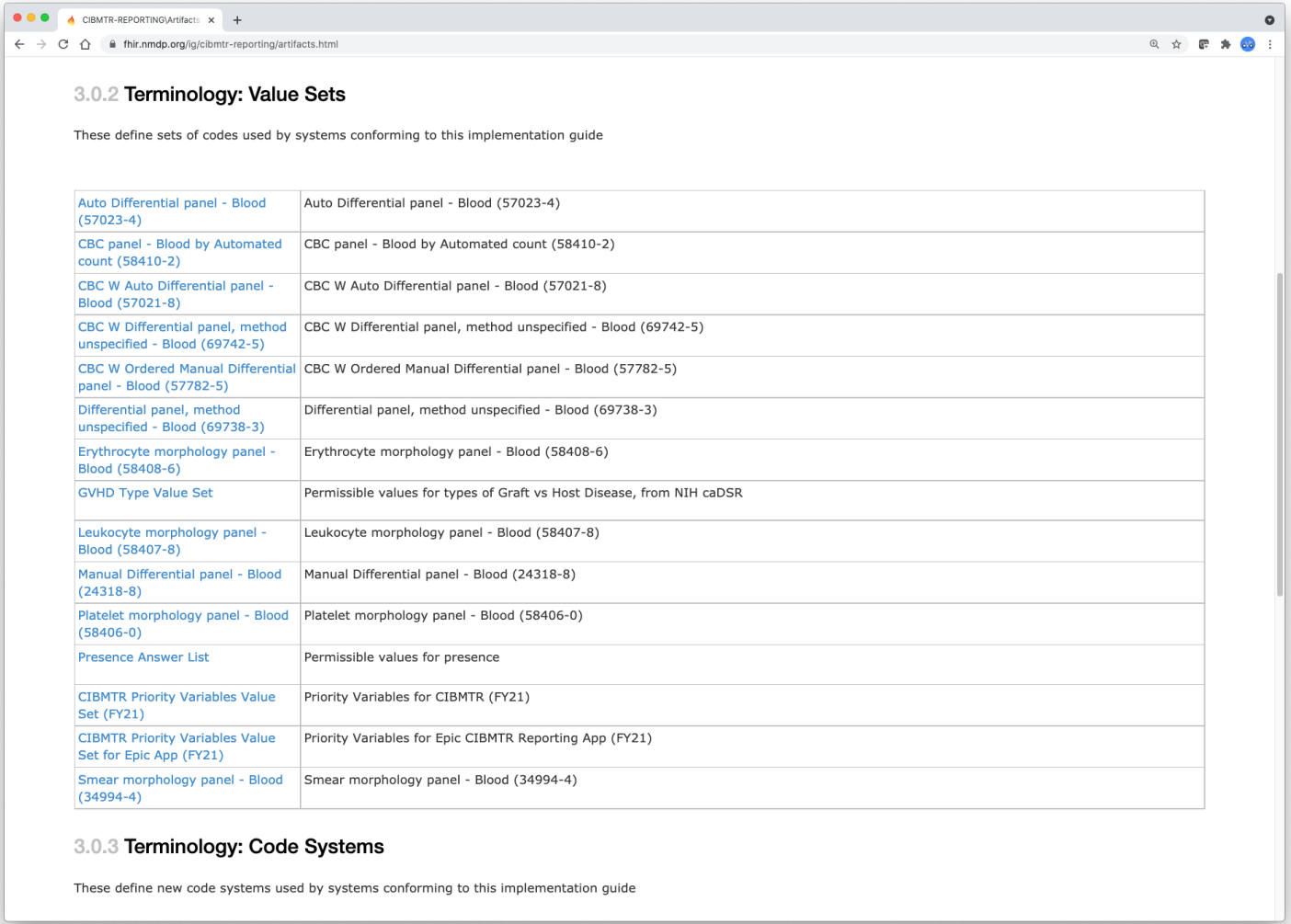
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1.2 CRID Lookup/Creation

A REST API is available for enabling users to lookup and/or create CRIDs based on patient demographic information. This is described in Direct FHIR for


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3 Artifacts Summary
This page provides a list of the FHIR artifacts defined as part of this implementation guide.
Contents:
Structures: Resource Profiles
Terminology: Value Sets
Terminology: Code Systems
Example: Example Instances
3.0.1 Structures: Resource Profiles
These define constraints on FHIR resources for systems conforming to this implementation guide
CIBMTR Observation Laboratory Results: Priority Variables
CIBMTR Observation Laboratory Results: Priority Variables
CIBMTR Observation Laboratory Results Profile (us-core)
CIBMTR Observation Laboratory Results Profile (us-core)
CIBMTR Patient Profile (us-core)
CIBMTR Patient Profile for CRID assignment (us-core)
3.0.2 Terminology: Value Sets
These define sets of codes used by systems conforming to this implementation guide
Auto Differential panel - Blood (57023-4)
Auto Differential panel - Blood (57023-4)



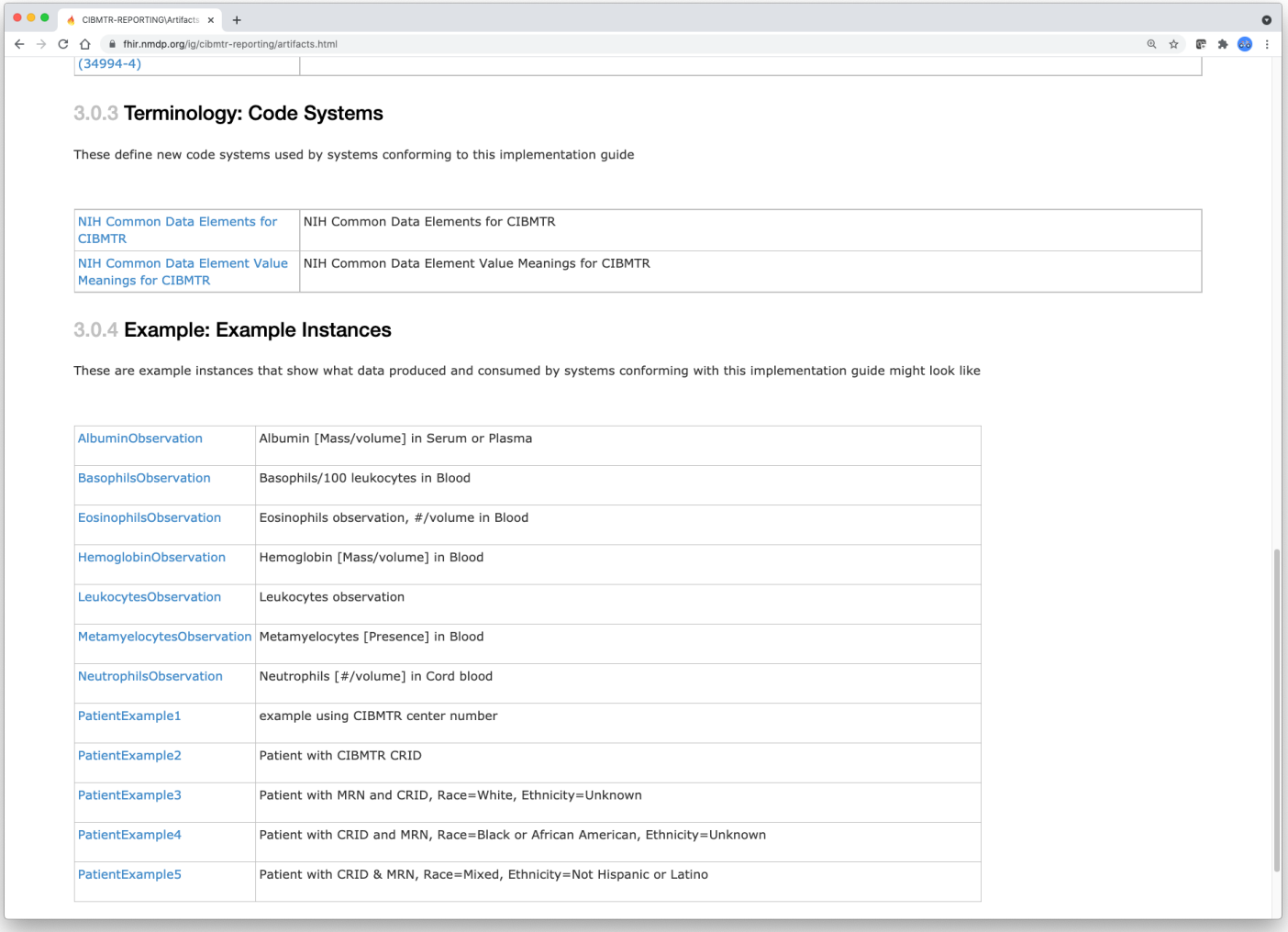
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Auto Differential panel - Blood (57023-4)	Auto Differential panel - Blood (57023-4)
CBC panel - Blood by Automated count (58410-2)	CBC panel - Blood by Automated count (58410-2)
CBC W Auto Differential panel - Blood (57021-8)	CBC W Auto Differential panel - Blood (57021-8)
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Differential panel, method unspecified - Blood (69738-3)	Differential panel, method unspecified - Blood (69738-3)
Erythrocyte morphology panel - Blood (58408-6)	Erythrocyte morphology panel - Blood (58408-6)
GVHD Type Value Set	Permissible values for types of Graft vs Host Disease, from NIH caDSR
Leukocyte morphology panel - Blood (58407-8)	Leukocyte morphology panel - Blood (58407-8)
Manual Differential panel - Blood (24318-8)	Manual Differential panel - Blood (24318-8)
Platelet morphology panel - Blood (58406-0)	Platelet morphology panel - Blood (58406-0)
Presence Answer List	Permissible values for presence
CIBMTR Priority Variables Value Set (FY21)	Priority Variables for CIBMTR (FY21)
CIBMTR Priority Variables Value Set for Epic App (FY21)	Priority Variables for Epic CIBMTR Reporting App (FY21)
Smear morphology panel - Blood (34994-4)	Smear morphology panel - Blood (34994-4)

3.0.3 Terminology: Code Systems

These define new code systems used by systems conforming to this implementation guide



(34994-4)	

3.0.3 Terminology: Code Systems

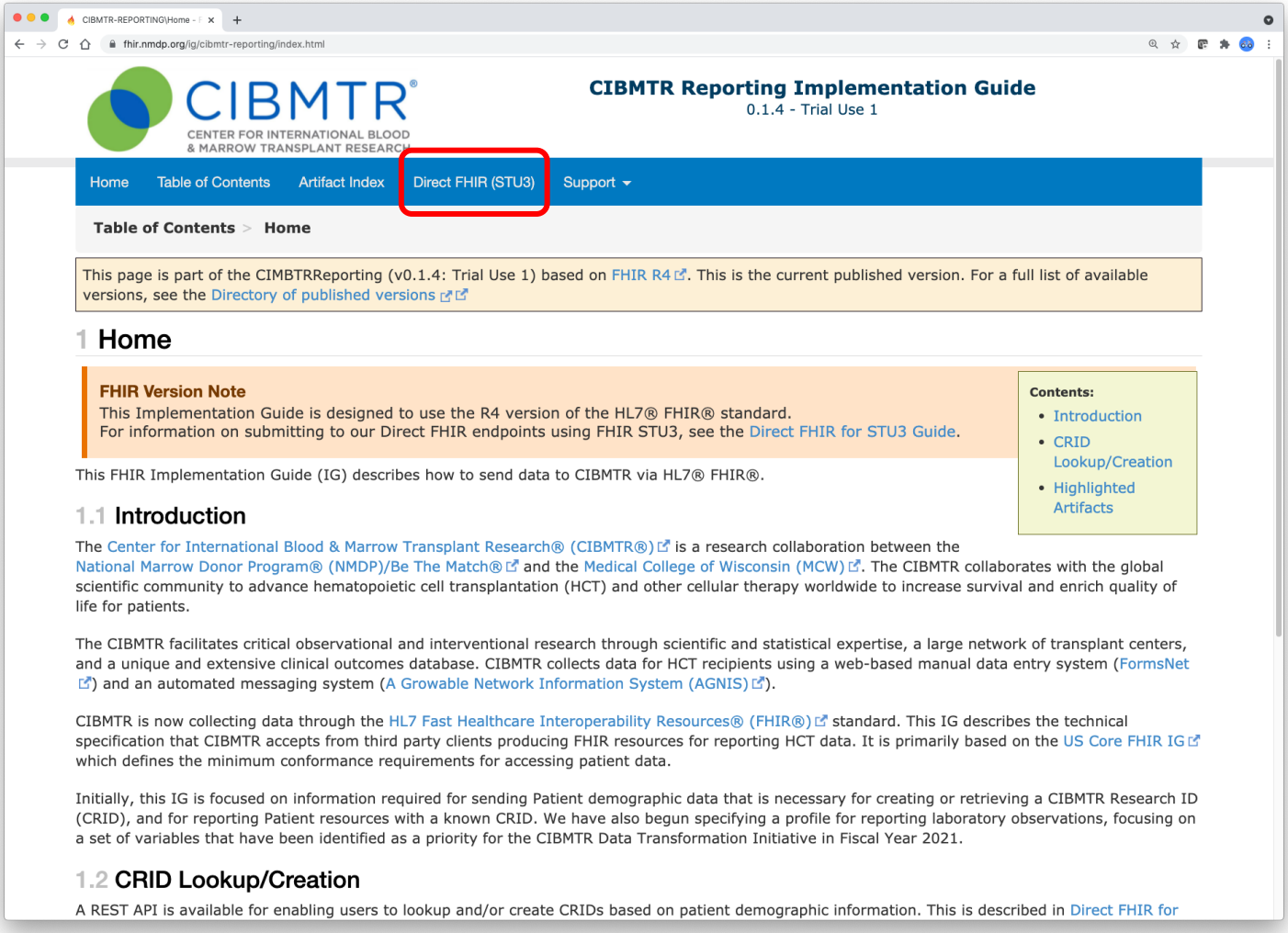
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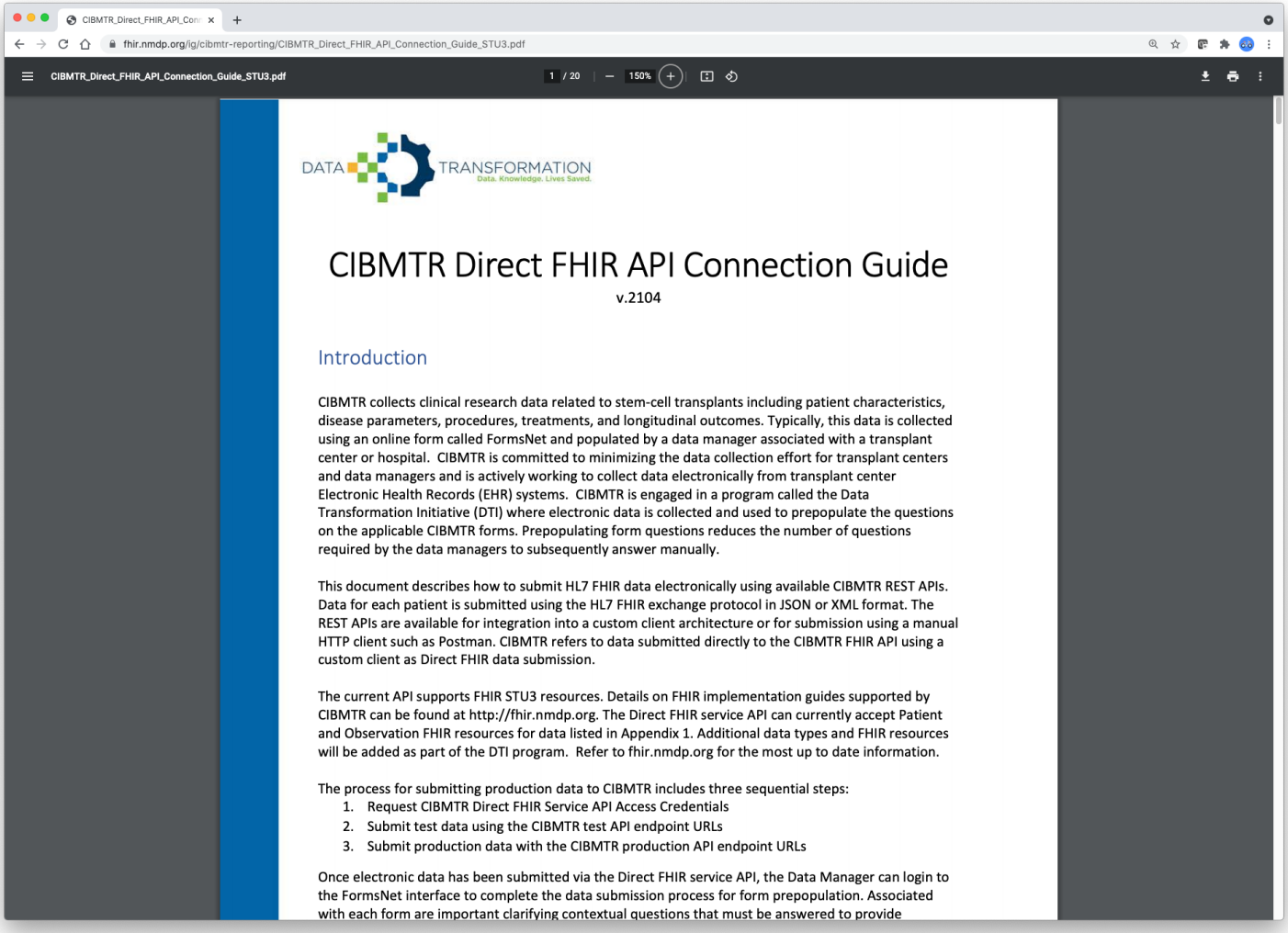
NIH Common Data Elements for CIBMTR	NIH Common Data Elements for CIBMTR
NIH Common Data Element Value Meanings for CIBMTR	NIH Common Data Element Value Meanings for CIBMTR

3.0.4 Example: Example Instances

These are example instances that show what data produced and consumed by systems conforming with this implementation guide might look like

AlbuminObservation	Albumin [Mass/volume] in Serum or Plasma
BasophilsObservation	Basophils/100 leukocytes in Blood
EosinophilsObservation	Eosinophils observation, #/volume in Blood
HemoglobinObservation	Hemoglobin [Mass/volume] in Blood
LeukocytesObservation	Leukocytes observation
MetamyelocytesObservation	Metamyelocytes [Presence] in Blood
NeutrophilsObservation	Neutrophils [#/volume] in Cord blood
PatientExample1	example using CIBMTR center number
PatientExample2	Patient with CIBMTR CRID
PatientExample3	Patient with MRN and CRID, Race=White, Ethnicity=Unknown
PatientExample4	Patient with CRID and MRN, Race=Black or African American, Ethnicity=Unknown
PatientExample5	Patient with CRID & MRN, Race=Mixed, Ethnicity=Not Hispanic or Latino


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CRID Lookup/Creation
Highlighted Artifacts
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CIBMTR Direct FHIR API Connection Guide

v.2104

Introduction

CIBMTR collects clinical research data related to stem-cell transplants including patient characteristics, disease parameters, procedures, treatments, and longitudinal outcomes. Typically, this data is collected using an online form called FormsNet and populated by a data manager associated with a transplant center or hospital. CIBMTR is committed to minimizing the data collection effort for transplant centers and data managers and is actively working to collect data electronically from transplant center Electronic Health Records (EHR) systems. CIBMTR is engaged in a program called the Data Transformation Initiative (DTI) where electronic data is collected and used to prepopulate the questions on the applicable CIBMTR forms. Prepopulating form questions reduces the number of questions required by the data managers to subsequently answer manually.

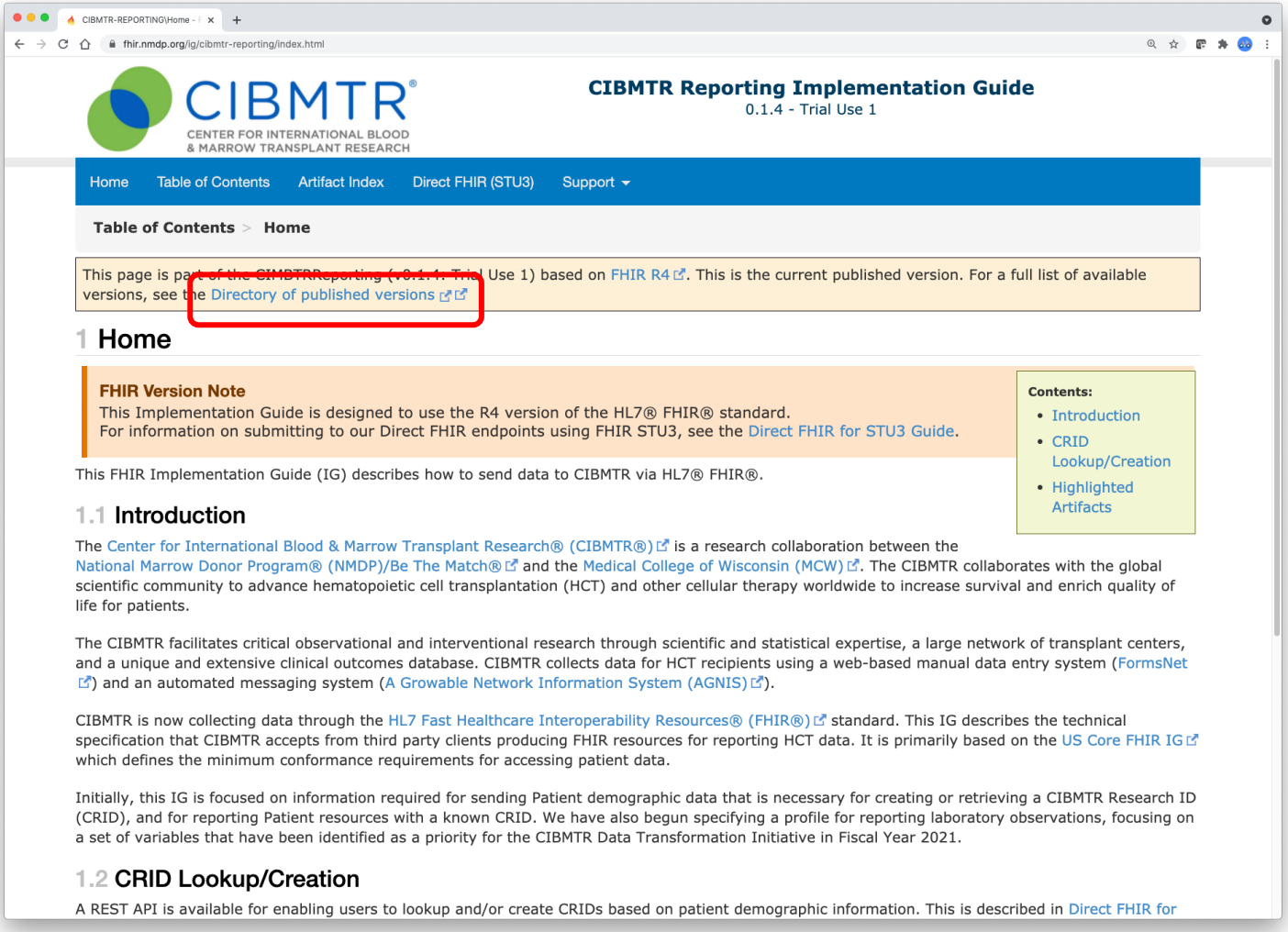
This document describes how to submit HL7 FHIR data electronically using available CIBMTR REST APIs. Data for each patient is submitted using the HL7 FHIR exchange protocol in JSON or XML format. The REST APIs are available for integration into a custom client architecture or for submission using a manual HTTP client such as Postman. CIBMTR refers to data submitted directly to the CIBMTR FHIR API using a custom client as Direct FHIR data submission.

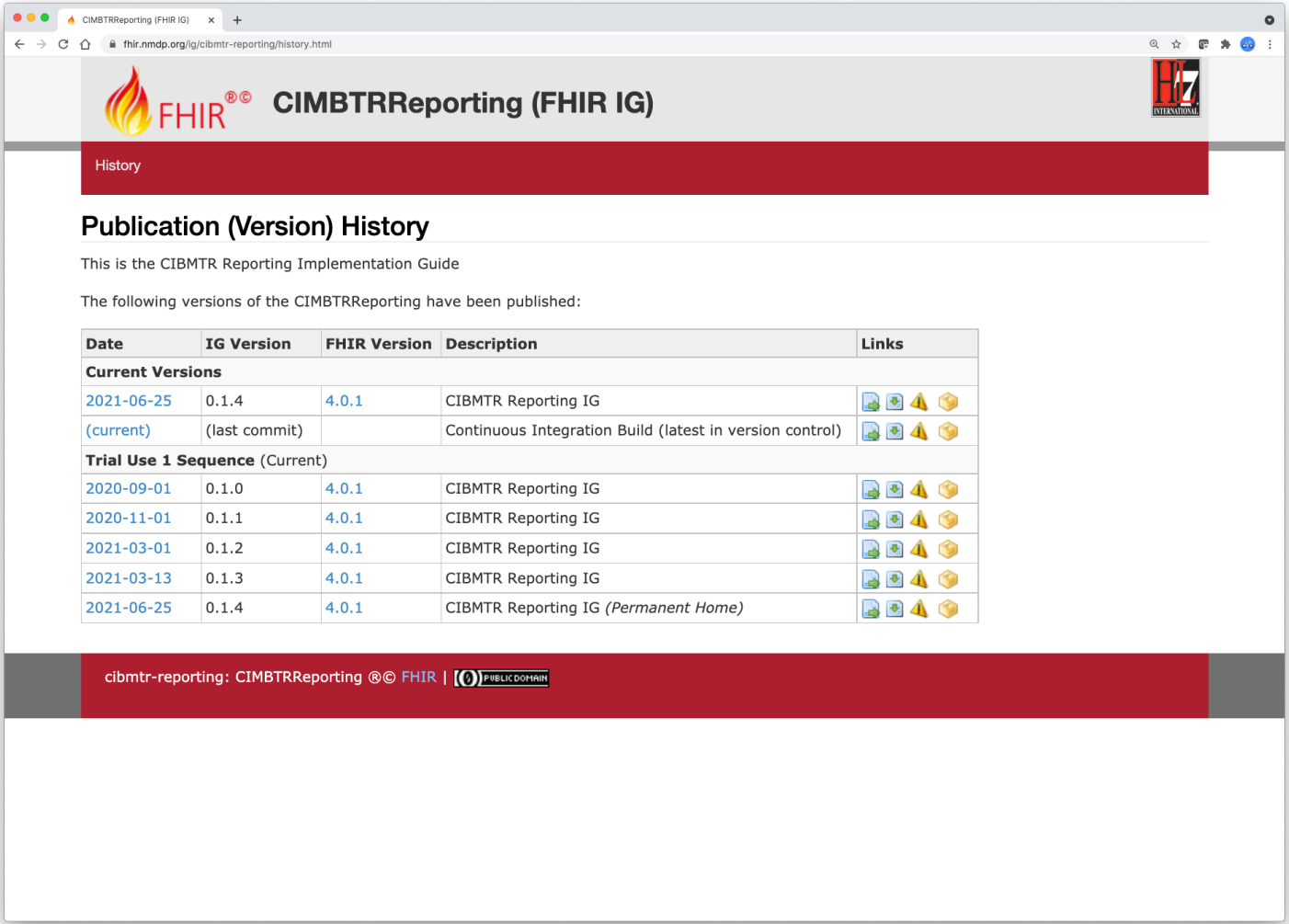
The current API supports FHIR STU3 resources. Details on FHIR implementation guides supported by CIBMTR can be found at http://fhir.nmdp.org. The Direct FHIR service API can currently accept Patient and Observation FHIR resources for data listed in Appendix 1. Additional data types and FHIR resources will be added as part of the DTI program. Refer to fhir.nmdp.org for the most up to date information.

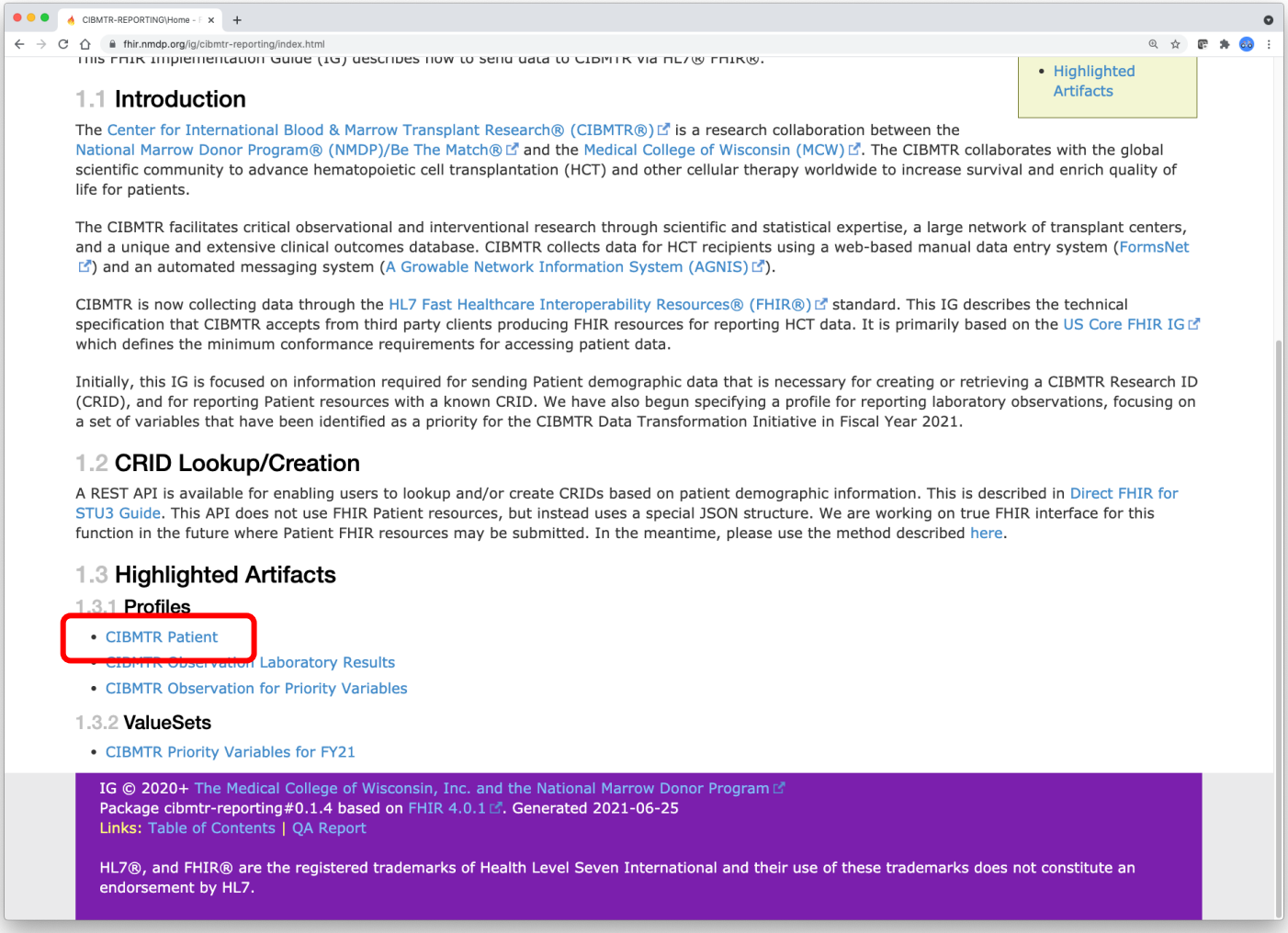
The process for submitting production data to CIBMTR includes three sequential steps:

1. Request CIBMTR Direct FHIR Service API Access Credentials
2. Submit test data using the CIBMTR test API endpoint URLs
3. Submit production data with the CIBMTR production API endpoint URLs

Once electronic data has been submitted via the Direct FHIR service API, the Data Manager can login to the FormsNet interface to complete the data submission process for form prepopulation. Associated with each form are important clarifying contextual questions that must be answered to provide


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CIMBTRReporting (FHIR IG)
History
Publication (Version) History
This is the CIBMTR Reporting Implementation Guide
The following versions of the CIMBTRReporting have been published:
Date | IG Version | FHIR Version | Description | Links
Current Versions
2021-06-25 | 0.1.4 | 4.0.1 | CIBMTR Reporting IG
(current) | (last commit) | | Continuous Integration Build (latest in version control)
Trial Use 1 Sequence (Current)
2020-09-01 | 0.1.0 | 4.0.1 | CIBMTR Reporting IG
2020-11-01 | 0.1.1 | 4.0.1 | CIBMTR Reporting IG
2021-03-01 | 0.1.2 | 4.0.1 | CIBMTR Reporting IG
2021-03-13 | 0.1.3 | 4.0.1 | CIBMTR Reporting IG
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Package cibmtr-reporting#0.1.4 based on FHIR 4.0.1. Generated 2021-06-25
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3.3.1 Resource Profile: CIBMTR Patient Profile (us-core)
Defining URL: http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-patient
Version: 0.1.4
Name: CIBMTRPatient
Title: CIBMTR Patient Profile (us-core)
Status: Active as of 2021-06-25T13:41:27-05:00
Definition: CIBMTR Patient Profile for CRID assignment (us-core)
Publisher: The Medical College of Wisconsin, Inc. and the National Marrow Donor Program
Source Resource: XML / JSON / Turtle
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-patient
Intro
This is the introduction to the CIBMTR Patient Profile. This is found at the beginning of the profile.
3.3.1.1 Formal Views of Profile Content
Description of Profiles, Differentials, Snapshots and how the different presentations work

3.3.1.1 Formal Views of Profile Content

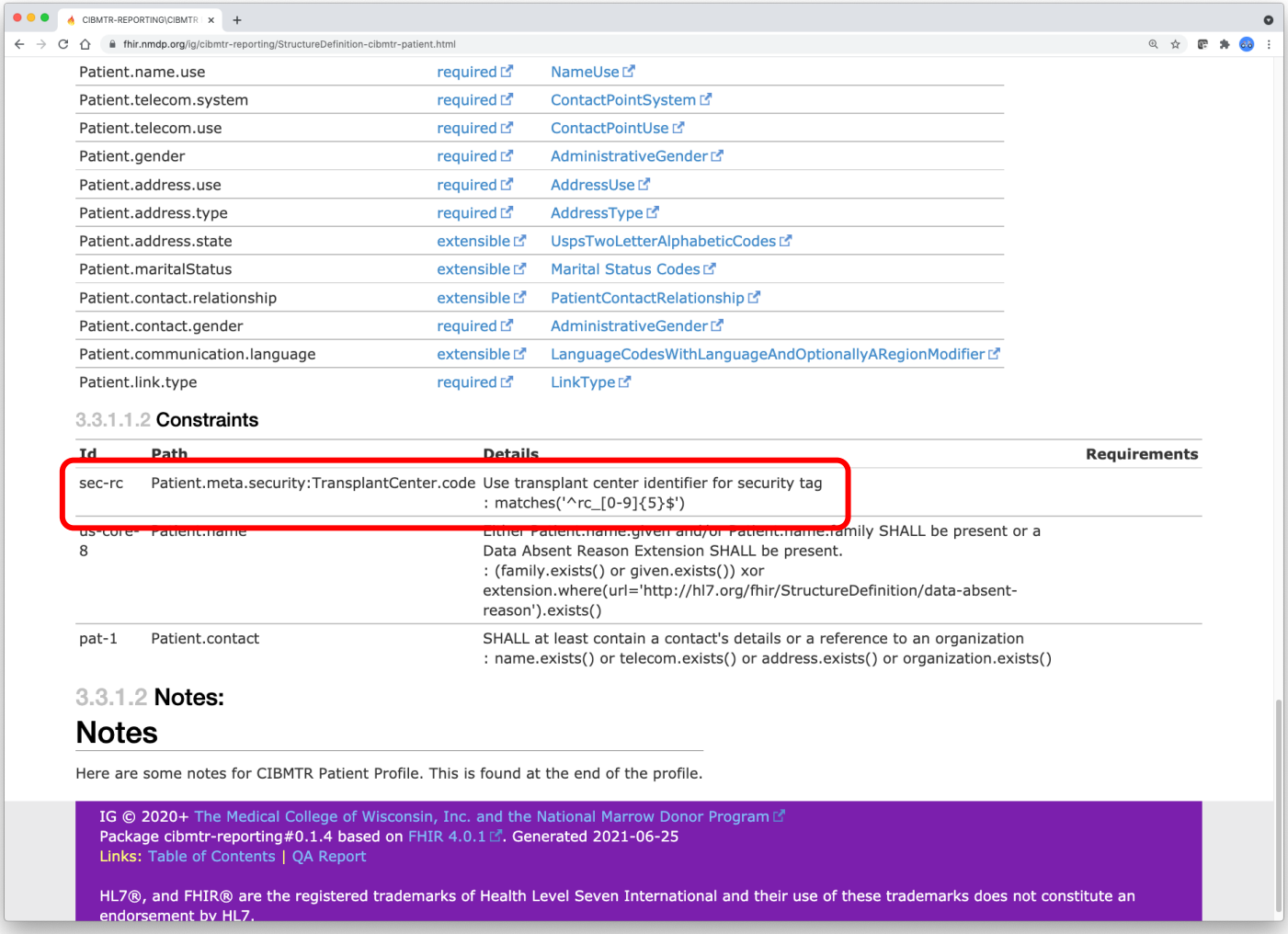
Description of Profiles, Differentials, Snapshots and how the different presentations work.

Text Summary | **Differential Table** | Snapshot Table | Snapshot Table (Must Support) | All

This structure is derived from USCorePatientProfile

Name	Flags	Card.	Type	Description & Constraints
Patient		0..*	USCorePatientProfile	Information about an individual or animal receiving health care services
meta				
security		1..*	*(Slice Definition)*	*Security Labels applied to this resource* ***Slice:*** *Unordered, Open by pattern:system*
security:TransplantCenter	S	1..1	Coding	Security Labels applied to this resource
system		1..1	uri	Identity of the terminology system **Required Pattern:** http://terminology.cibmtr.org/codesystem/transplant-center
code		1..1	code	Symbol in syntax defined by the system **sec-rc:** Use transplant center identifier for security tag
	S		string	Mother's Maiden Name **URL:** http://hl7.org/fhir/StructureDefinition/patient-mothersMaidenName
us-core-birthsex	S	0..1	code	Extension **URL:** http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex **Binding:** Birth Sex (required): Code for sex assigned at birth
patient-birthPlace	S	0..1	Address	Place of Birth for patient **URL:** http://hl7.org/fhir/StructureDefinition/patient-birthPlace
identifier		1..*	*(Slice Definition)*	*An identifier for this patient* ***Slice:*** *Unordered, Open by pattern:system*
identifier:CRID	S	0..1	Identifier	An identifier for this patient
system		1..1	uri	The namespace for the identifier value **Required Pattern:** http://terminology.cibmtr.org/identifier/CRID
identifier:GRID	S	0..1	Identifier	An identifier for this patient
type				
coding				
system		0..1	uri	Identity of the terminology system **Required Pattern:** https://www.hl7.org/fhir/v2/0203/index.html
code		0..1	code	Symbol in syntax defined by the system **Required Pattern:** DR
system		1..1	uri	The namespace for the identifier value **Required Pattern:** http://www.isbt128.org/uri/GRID
identifier:NMDPRecipient	S	0..1	Identifier	An identifier for this patient
system		1..1	uri	The namespace for the identifier value **Required Pattern:** http://terminology.nmdp.org/identifier/recipient
identifier:CIBMTR-non-NMDP-donor		0..1	Identifier	An identifier for this patient
system		1..1	uri	The namespace for the identifier value **Required Pattern:** http://terminology.cibmtr.org/identifier/non-nmdp-donor
identifier:SSN	S	0..1	Identifier	An identifier for this patient
system		1..1	uri	The namespace for the identifier value **Required Pattern:** http://hl7.org/fhir/sid/us-ssn





Patient.name.use	required	NameUse
Patient.telecom.system	required	ContactPointSystem
Patient.telecom.use	required	ContactPointUse
Patient.gender	required	AdministrativeGender
Patient.address.use	required	AddressUse
Patient.address.type	required	AddressType
Patient.address.state	extensible	UspsTwoLetterAlphabeticCodes
Patient.maritalStatus	extensible	Marital Status Codes
Patient.contact.relationship	extensible	PatientContactRelationship
Patient.contact.gender	required	AdministrativeGender
Patient.communication.language	extensible	LanguageCodesWithLanguageAndOptionallyARegionModifier
Patient.link.type	required	LinkType

3.3.1.1.2 Constraints

Id	Path	Details	Requirements
sec-rc	Patient.meta.security:TransplantCenter.code	Use transplant center identifier for security tag : matches('^rc_[0-9]{5}$')	
us-core-8	Patient.name	Either Patient.name.given and/or Patient.name.family SHALL be present or a Data Absent Reason Extension SHALL be present. : (family.exists() or given.exists()) xor extension.where(url='http://hl7.org/fhir/StructureDefinition/data-absent-reason').exists()	
pat-1	Patient.contact	SHALL at least contain a contact's details or a reference to an organization : name.exists() or telecom.exists() or address.exists() or organization.exists()	

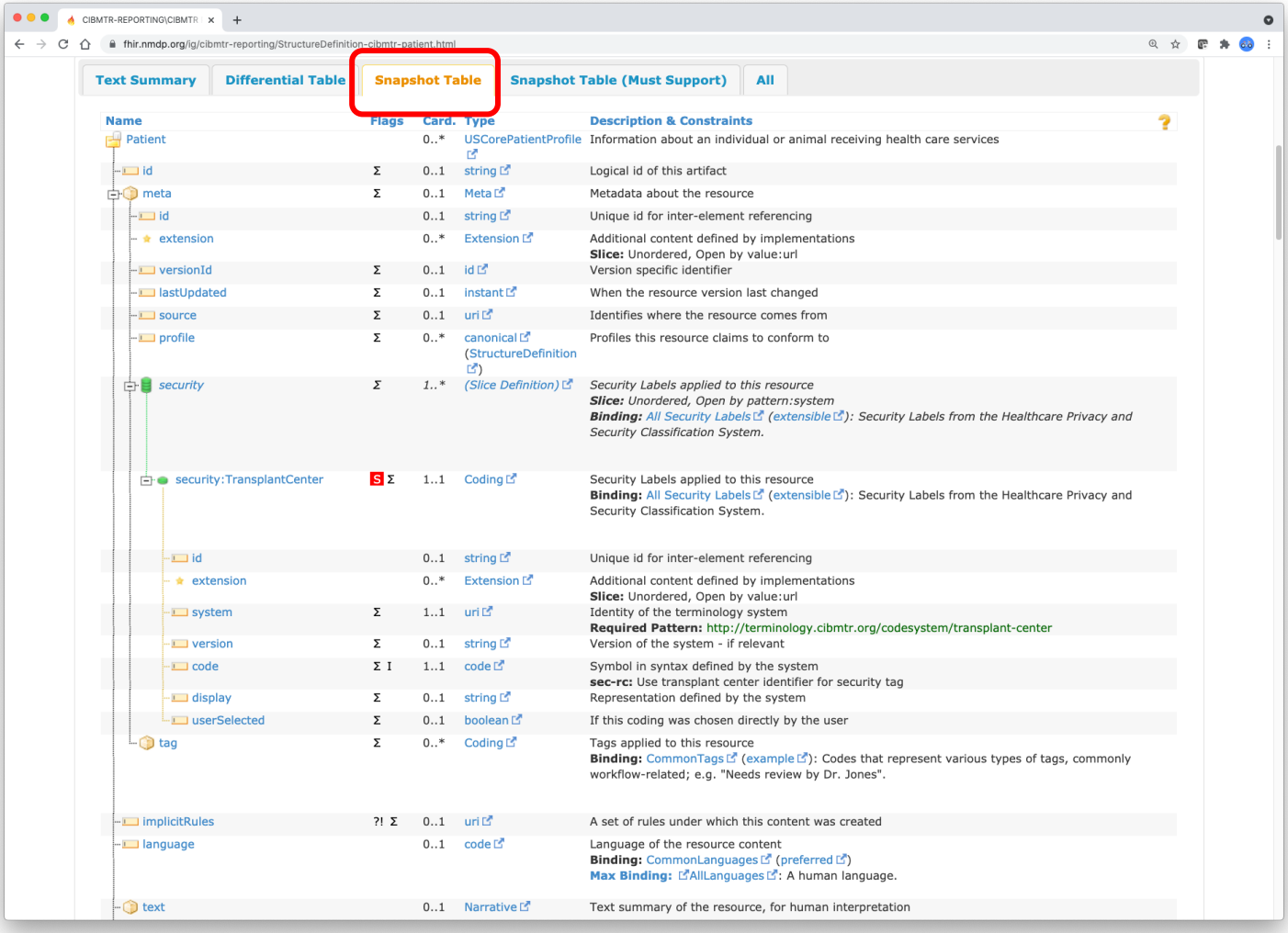
3.3.1.2 Notes:

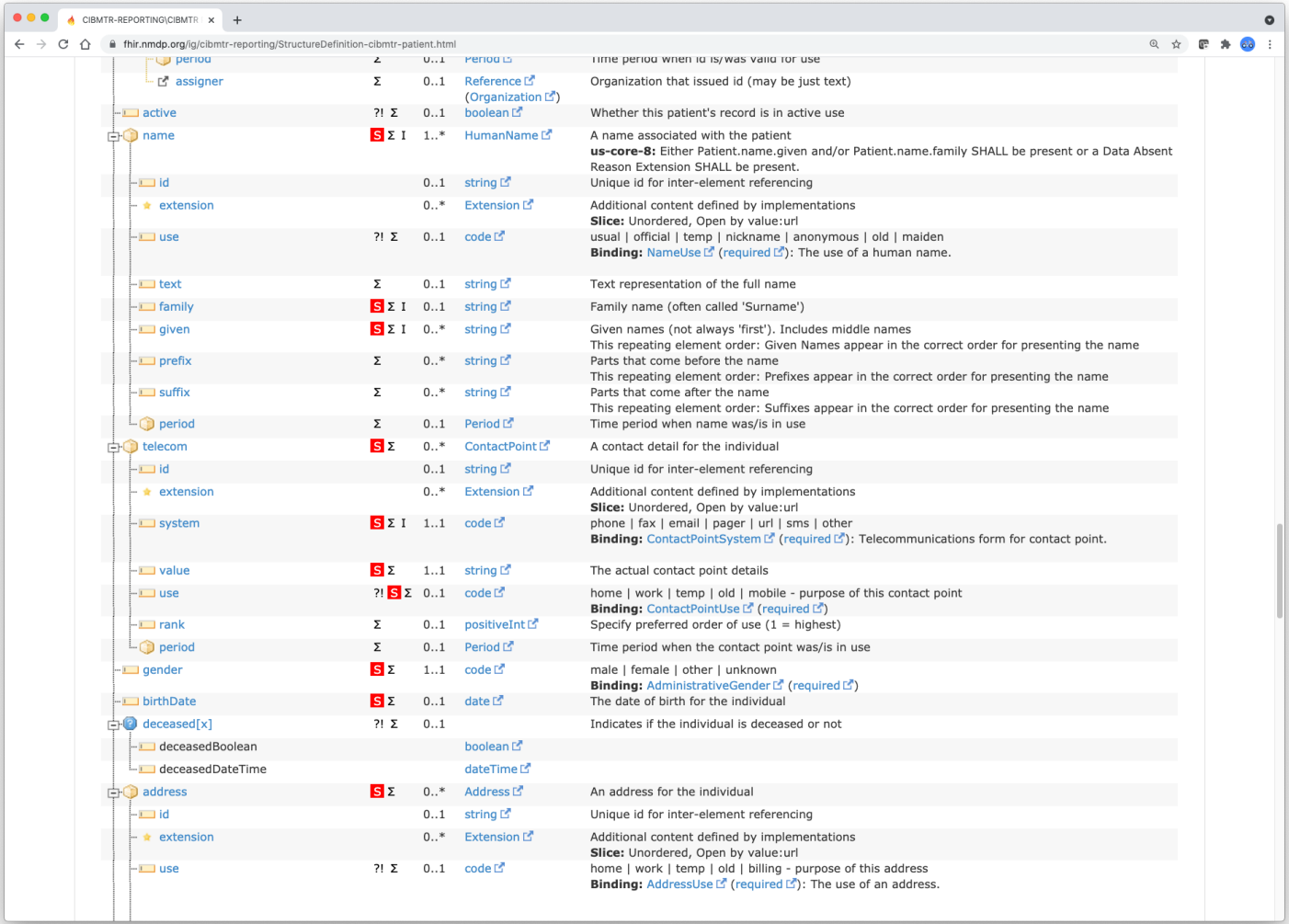
Notes

Here are some notes for CIBMTR Patient Profile. This is found at the end of the profile.

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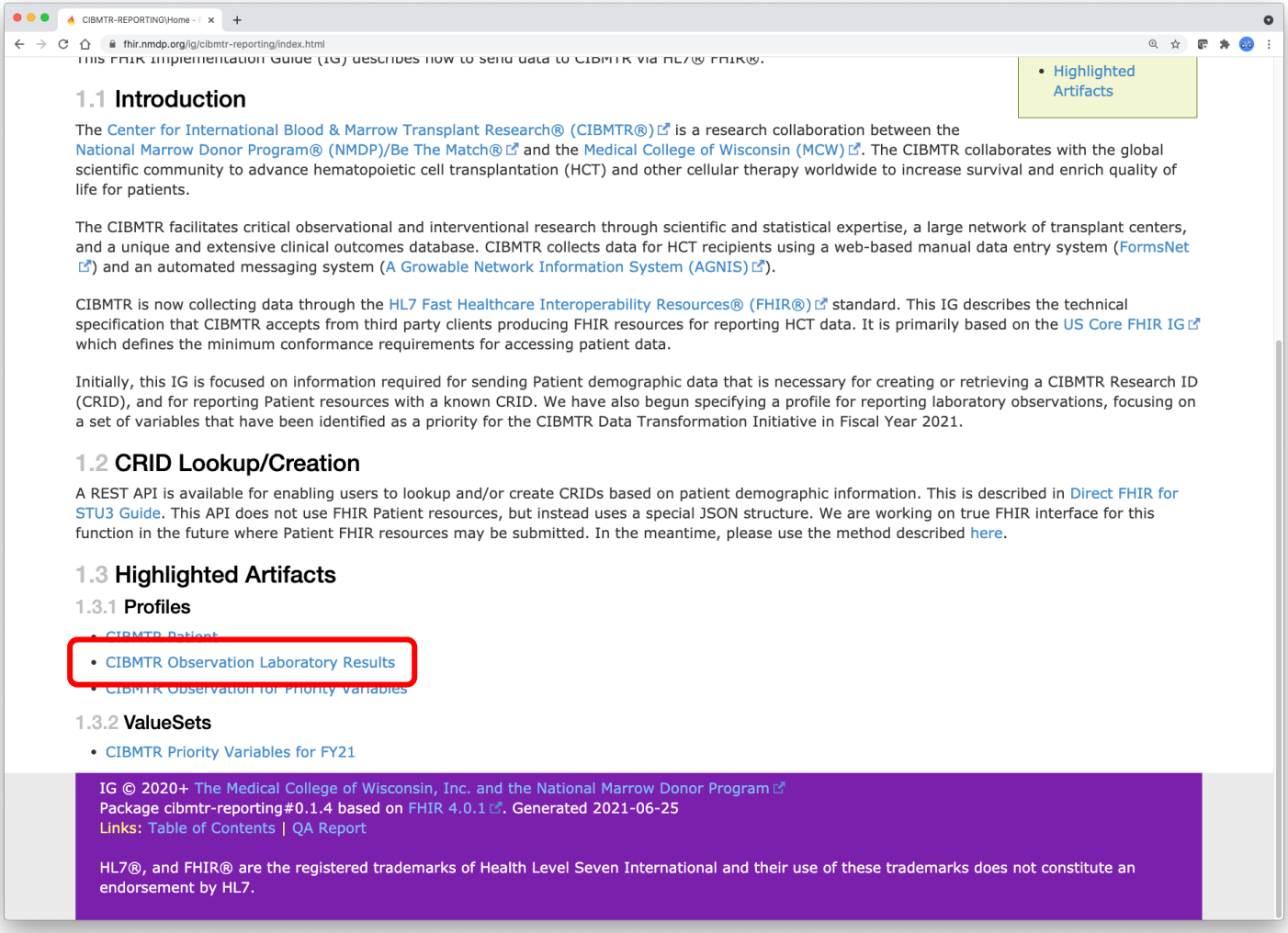

CIBMTR-REPORTING\CIBMTR
fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-patient.html
Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
Name
Flags
Card.
Type
Description & Constraints
Patient
0..*
USCorePatientProfile
Information about an individual or animal receiving health care services
id
Σ
0..1
string
Logical id of this artifact
meta
Σ
0..1
Meta
Metadata about the resource
id
0..1
string
Unique id for inter-element referencing
extension
0..*
Extension
Additional content defined by implementations
Slice: Unordered, Open by value:url
versionId
Σ
0..1
id
Version specific identifier
lastUpdated
Σ
0..1
instant
When the resource version last changed
source
Σ
0..1
uri
Identifies where the resource comes from
profile
Σ
0..*
canonical (StructureDefinition)
Profiles this resource claims to conform to
security
Σ
1..*
(Slice Definition)
Security Labels applied to this resource
Slice: Unordered, Open by pattern:system
Binding: All Security Labels (extensible): Security Labels from the Healthcare Privacy and Security Classification System.
security:TransplantCenter
S Σ
1..1
Coding
Security Labels applied to this resource
Binding: All Security Labels (extensible): Security Labels from the Healthcare Privacy and Security Classification System.
id
0..1
string
Unique id for inter-element referencing
extension
0..*
Extension
Additional content defined by implementations
Slice: Unordered, Open by value:url
system
Σ
1..1
uri
Identity of the terminology system
Required Pattern: http://terminology.cibmtr.org/codesystem/transplant-center
version
Σ
0..1
string
Version of the system - if relevant
code
Σ I
1..1
code
Symbol in syntax defined by the system
sec-rc: Use transplant center identifier for security tag
display
Σ
0..1
string
Representation defined by the system
userSelected
Σ
0..1
boolean
If this coding was chosen directly by the user
tag
Σ
0..*
Coding
Tags applied to this resource
Binding: CommonTags (example): Codes that represent various types of tags, commonly workflow-related; e.g. "Needs review by Dr. Jones".
implicitRules
?! Σ
0..1
uri
A set of rules under which this content was created
language
0..1
code
Language of the resource content
Binding: CommonLanguages (preferred)
Max Binding: AllLanguages: A human language.
text
0..1
Narrative
Text summary of the resource, for human interpretation

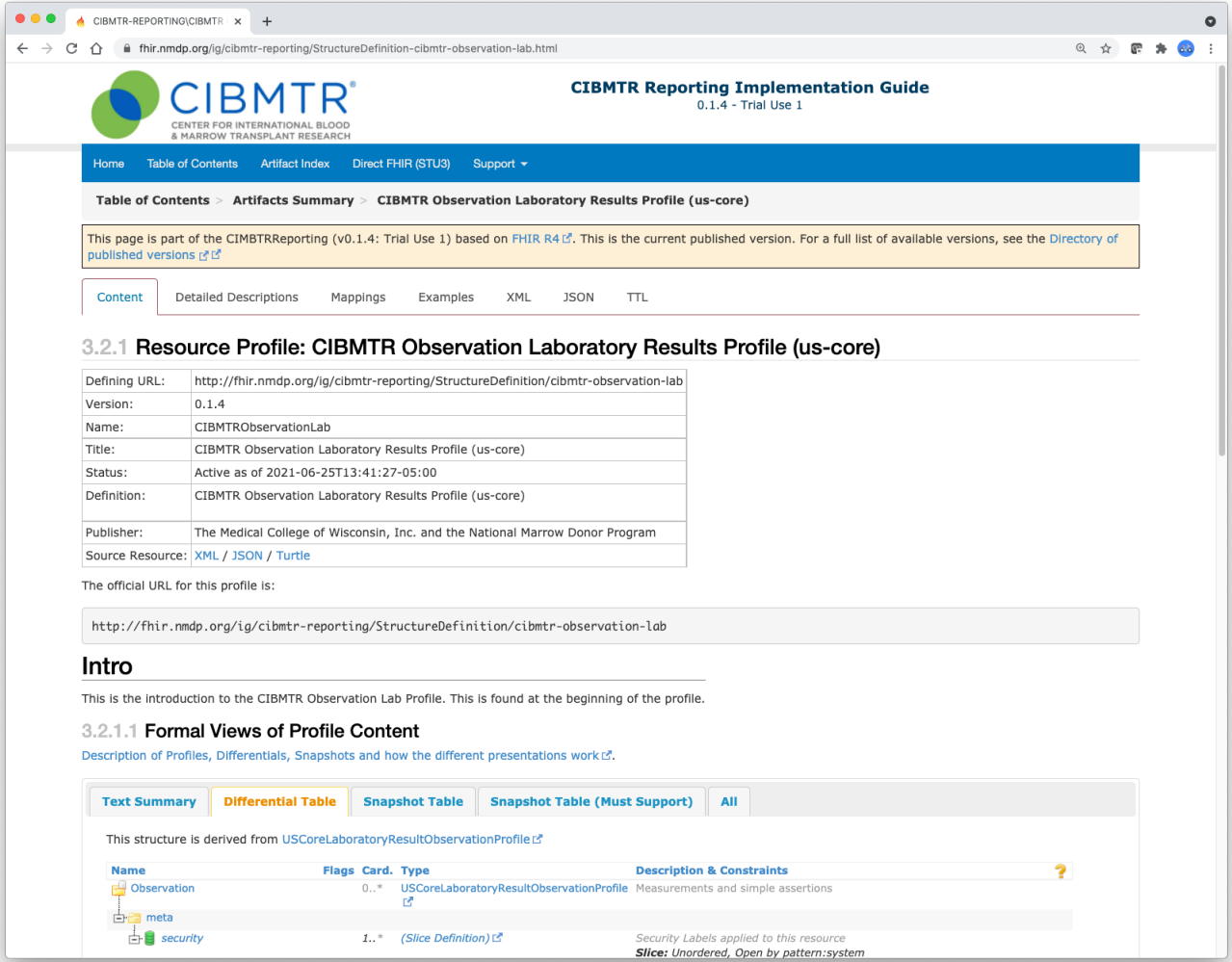


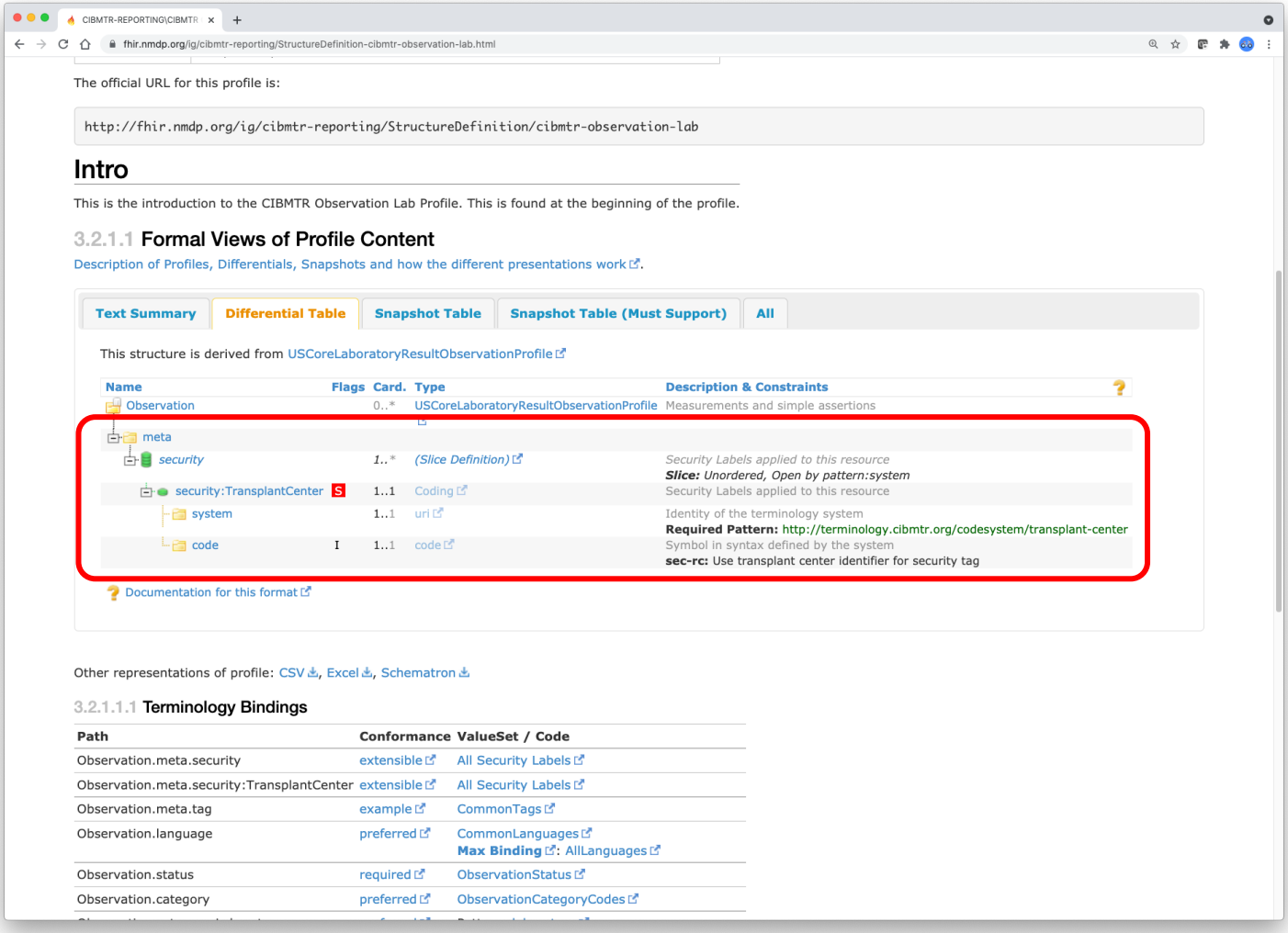
CIBMTR-REPORTING\CIBMTR

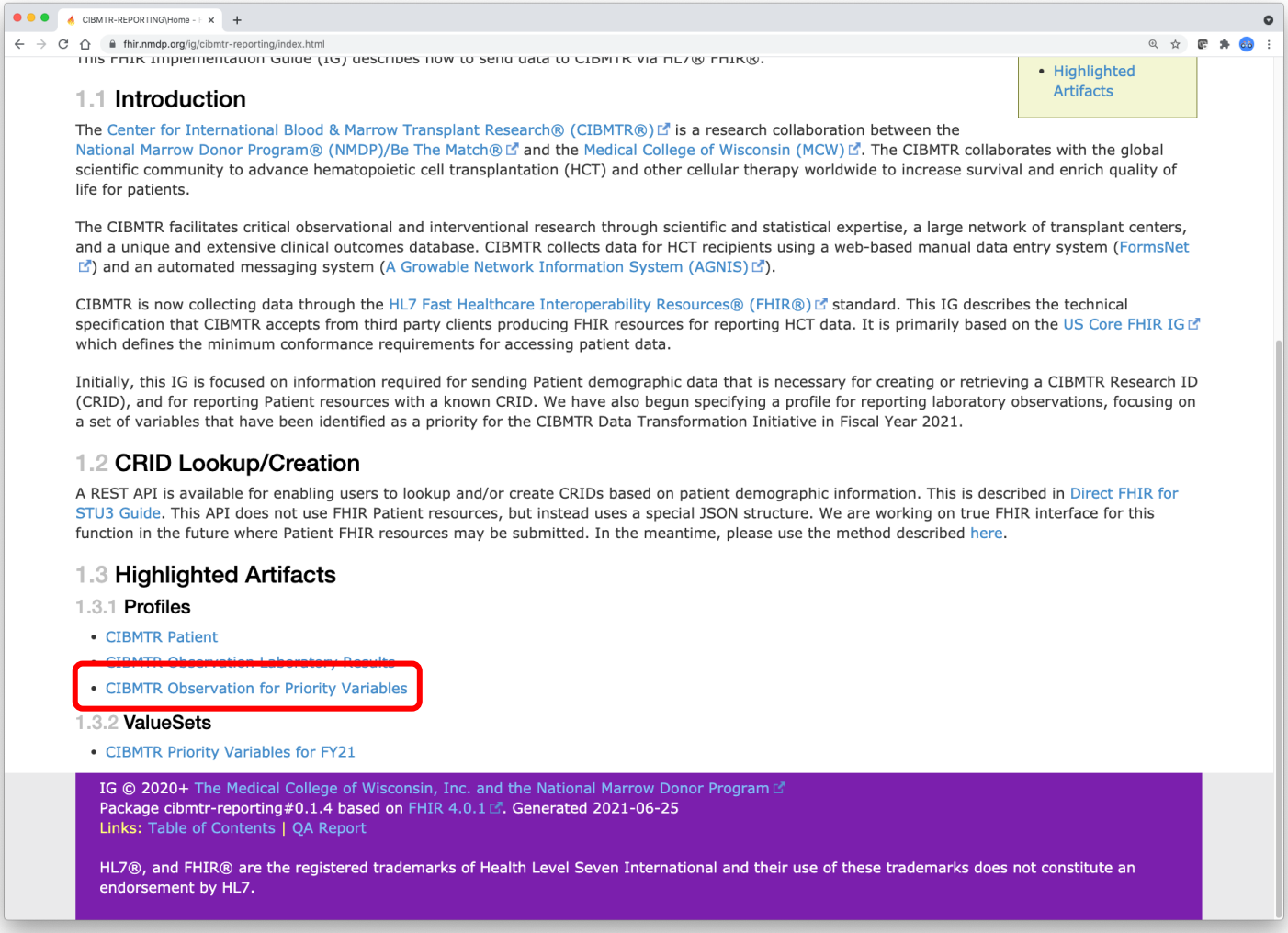
fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-patient.html

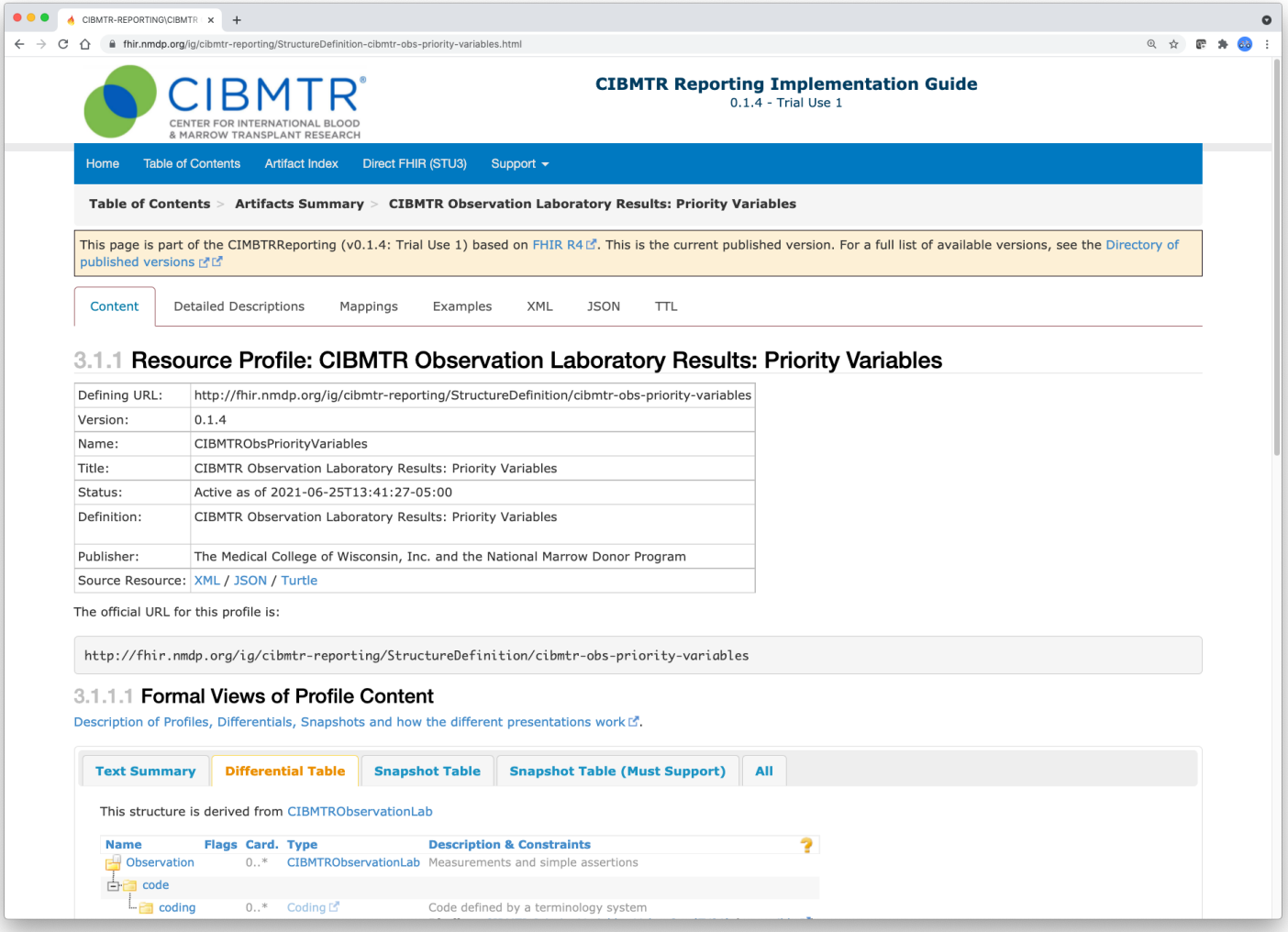
period	Σ	0..1	Period	Time period when id is/was valid for use
assigner	Σ	0..1	Reference (Organization)	Organization that issued id (may be just text)
active	?! Σ	0..1	boolean	Whether this patient's record is in active use
name	S Σ I	1..*	HumanName	A name associated with the patient **us-core-8:** Either Patient.name.given and/or Patient.name.family SHALL be present or a Data Absent Reason Extension SHALL be present.
id		0..1	string	Unique id for inter-element referencing
extension		0..*	Extension	Additional content defined by implementations **Slice:** Unordered, Open by value:url
use	?! Σ	0..1	code	usual | official | temp | nickname | anonymous | old | maiden **Binding:** NameUse (required): The use of a human name.
text	Σ	0..1	string	Text representation of the full name
family	S Σ I	0..1	string	Family name (often called 'Surname')
given	S Σ I	0..*	string	Given names (not always 'first'). Includes middle names This repeating element order: Given Names appear in the correct order for presenting the name
prefix	Σ	0..*	string	Parts that come before the name This repeating element order: Prefixes appear in the correct order for presenting the name
suffix	Σ	0..*	string	Parts that come after the name This repeating element order: Suffixes appear in the correct order for presenting the name
period	Σ	0..1	Period	Time period when name was/is in use
telecom	S Σ	0..*	ContactPoint	A contact detail for the individual
id		0..1	string	Unique id for inter-element referencing
extension		0..*	Extension	Additional content defined by implementations **Slice:** Unordered, Open by value:url
system	S Σ I	1..1	code	phone | fax | email | pager | url | sms | other **Binding:** ContactPointSystem (required): Telecommunications form for contact point.
value	S Σ	1..1	string	The actual contact point details
use	?! S Σ	0..1	code	home | work | temp | old | mobile - purpose of this contact point **Binding:** ContactPointUse (required)
rank	Σ	0..1	positiveInt	Specify preferred order of use (1 = highest)
period	Σ	0..1	Period	Time period when the contact point was/is in use
gender	S Σ	1..1	code	male | female | other | unknown **Binding:** AdministrativeGender (required)
birthDate	S Σ	0..1	date	The date of birth for the individual
deceased[x]	?! Σ	0..1		Indicates if the individual is deceased or not
deceasedBoolean			boolean	
deceasedDateTime			dateTime	
address	S Σ	0..*	Address	An address for the individual
id		0..1	string	Unique id for inter-element referencing
extension		0..*	Extension	Additional content defined by implementations **Slice:** Unordered, Open by value:url
use	?! Σ	0..1	code	home | work | temp | old | billing - purpose of this address **Binding:** AddressUse (required): The use of an address.

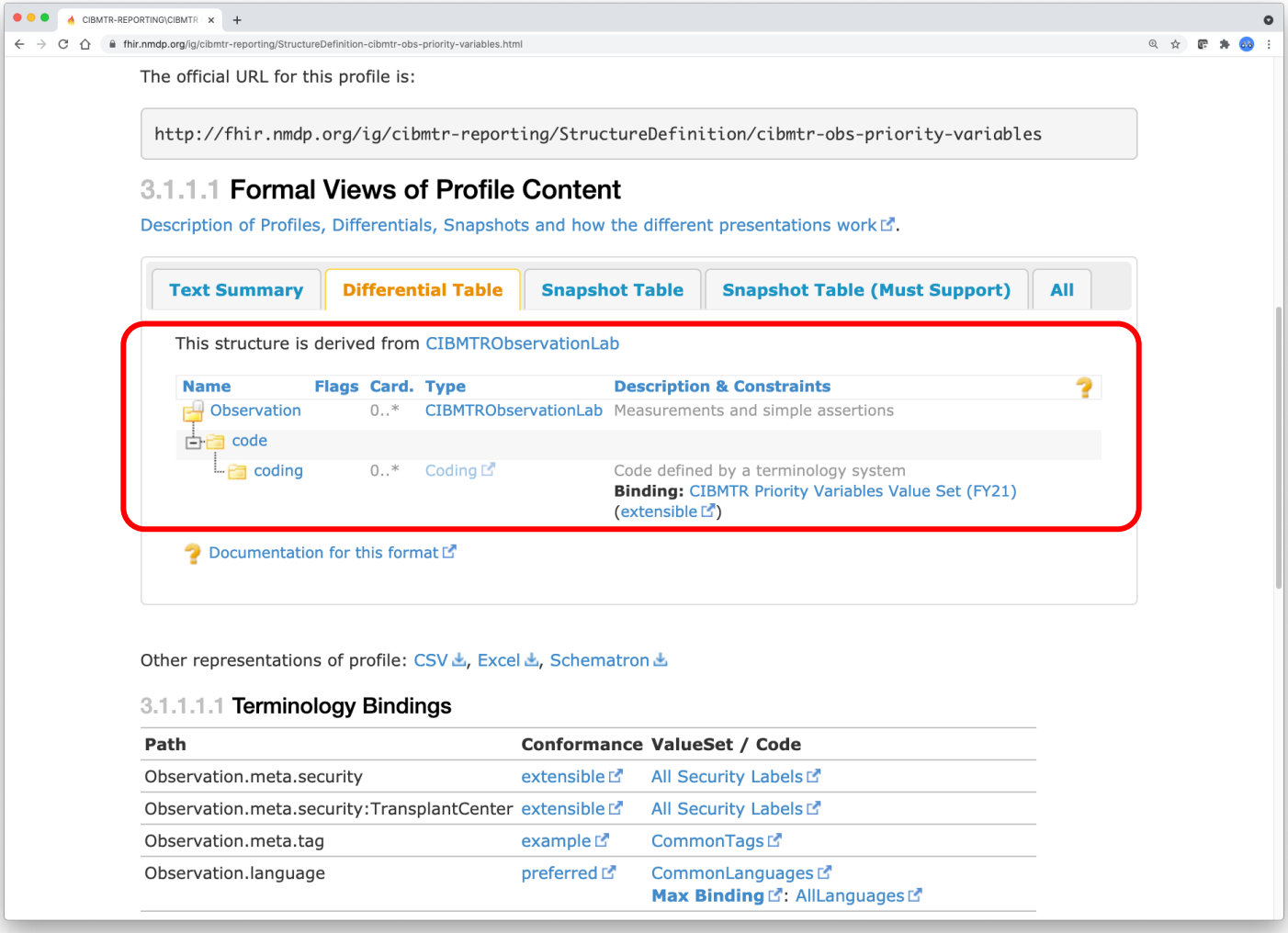

CIBMTR-REPORTING\Home -
fhir.nmdp.org/ig/cibmtr-reporting/index.html
This FHIR Implementation Guide (IG) describes how to send data to CIBMTR via HL7® FHIR®.
Highlighted Artifacts
1.1 Introduction
The Center for International Blood & Marrow Transplant Research® (CIBMTR®) is a research collaboration between the National Marrow Donor Program® (NMDP)/Be The Match® and the Medical College of Wisconsin (MCW). The CIBMTR collaborates with the global scientific community to advance hematopoietic cell transplantation (HCT) and other cellular therapy worldwide to increase survival and enrich quality of life for patients.
The CIBMTR facilitates critical observational and interventional research through scientific and statistical expertise, a large network of transplant centers, and a unique and extensive clinical outcomes database. CIBMTR collects data for HCT recipients using a web-based manual data entry system (FormsNet) and an automated messaging system (A Growable Network Information System (AGNIS)).
CIBMTR is now collecting data through the HL7 Fast Healthcare Interoperability Resources® (FHIR®) standard. This IG describes the technical specification that CIBMTR accepts from third party clients producing FHIR resources for reporting HCT data. It is primarily based on the US Core FHIR IG which defines the minimum conformance requirements for accessing patient data.
Initially, this IG is focused on information required for sending Patient demographic data that is necessary for creating or retrieving a CIBMTR Research ID (CRID), and for reporting Patient resources with a known CRID. We have also begun specifying a profile for reporting laboratory observations, focusing on a set of variables that have been identified as a priority for the CIBMTR Data Transformation Initiative in Fiscal Year 2021.
1.2 CRID Lookup/Creation
A REST API is available for enabling users to lookup and/or create CRIDs based on patient demographic information. This is described in Direct FHIR for STU3 Guide. This API does not use FHIR Patient resources, but instead uses a special JSON structure. We are working on true FHIR interface for this function in the future where Patient FHIR resources may be submitted. In the meantime, please use the method described here.
1.3 Highlighted Artifacts
1.3.1 Profiles
CIBMTR Patient
CIBMTR Observation Laboratory Results
CIBMTR Observation for Priority Variables
1.3.2 ValueSets
CIBMTR Priority Variables for FY21
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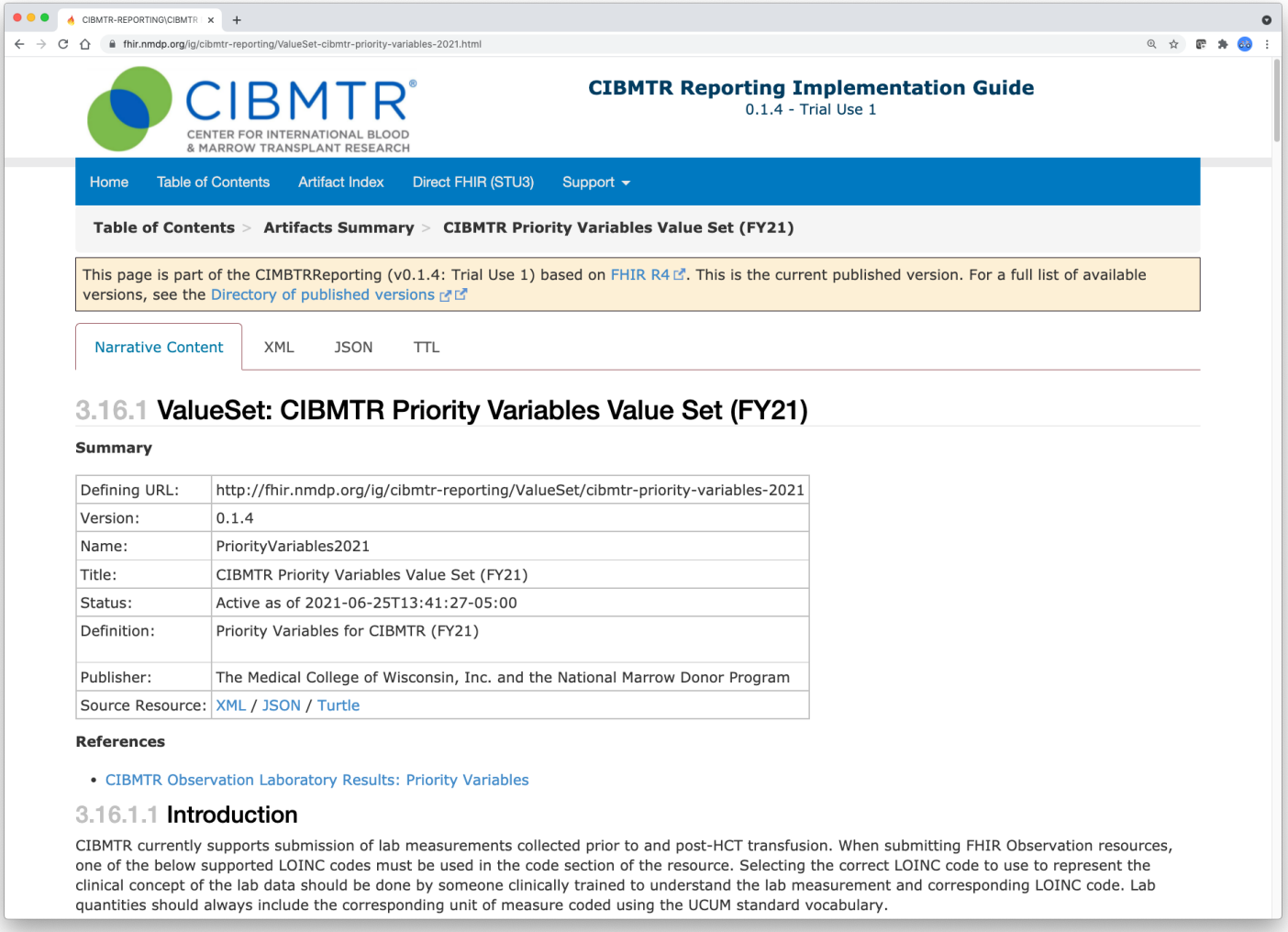

CIBMTR-REPORTING\CIBMTR
fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-observation-lab.html
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JSON
TTL
3.2.1 Resource Profile: CIBMTR Observation Laboratory Results Profile (us-core)
Defining URL: http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-observation-lab
Version: 0.1.4
Name: CIBMTRObservationLab
Title: CIBMTR Observation Laboratory Results Profile (us-core)
Status: Active as of 2021-06-25T13:41:27-05:00
Definition: CIBMTR Observation Laboratory Results Profile (us-core)
Publisher: The Medical College of Wisconsin, Inc. and the National Marrow Donor Program
Source Resource: XML / JSON / Turtle
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-observation-lab
Intro
This is the introduction to the CIBMTR Observation Lab Profile. This is found at the beginning of the profile.
3.2.1.1 Formal Views of Profile Content
Description of Profiles, Differentials, Snapshots and how the different presentations work.
Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
This structure is derived from USCoreLaboratoryResultObservationProfile
Name
Flags
Card.
Type
Description & Constraints
Observation
0..*
USCoreLaboratoryResultObservationProfile
Measurements and simple assertions
meta
security
1..*
(Slice Definition)
Security Labels applied to this resource
Slice: Unordered, Open by pattern:system

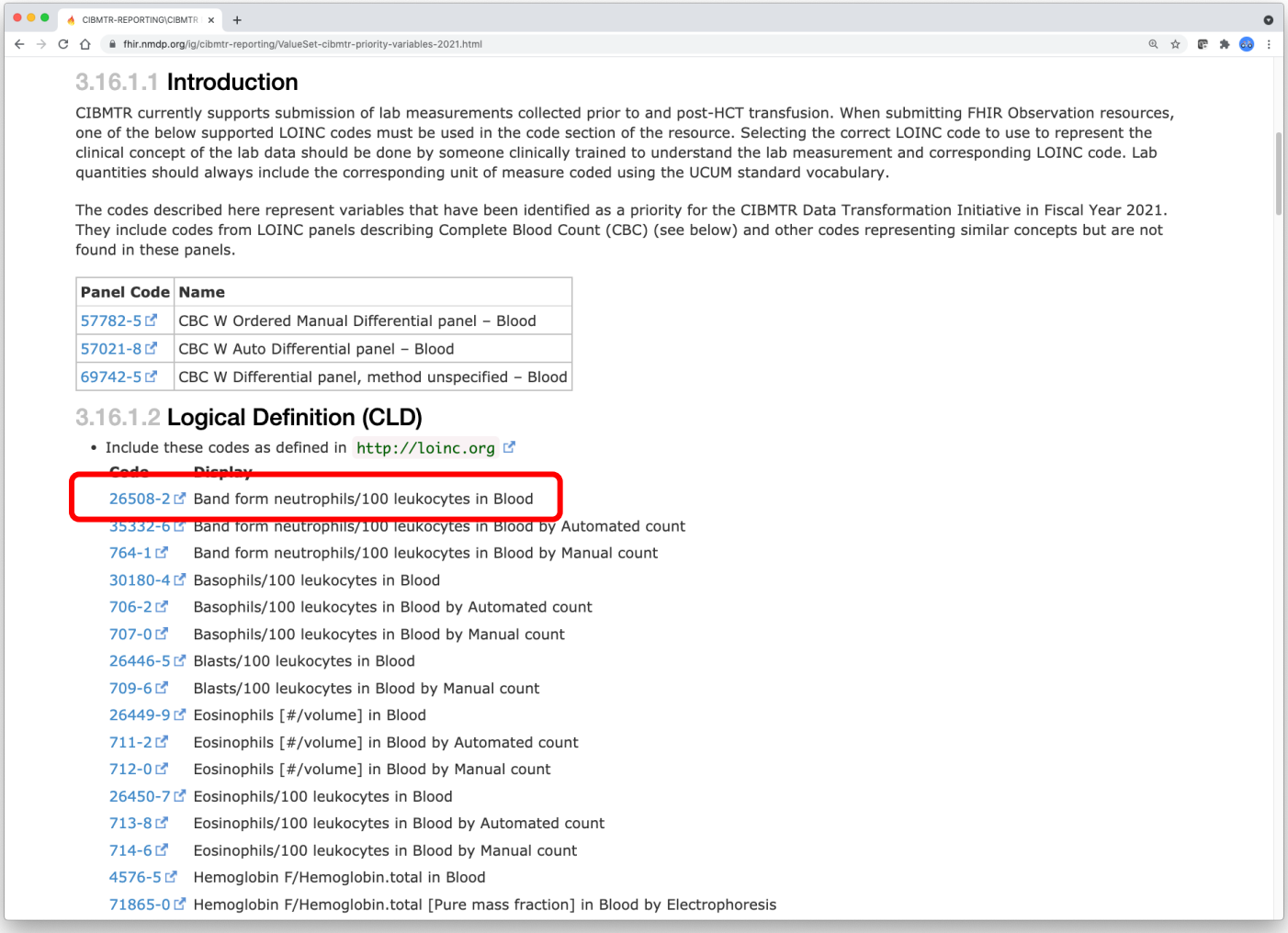

CIBMTR-REPORTING\CIBMTR
fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-observation-lab.html
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-observation-lab
Intro
This is the introduction to the CIBMTR Observation Lab Profile. This is found at the beginning of the profile.
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Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
This structure is derived from USCoreLaboratoryResultObservationProfile
Name
Flags
Card.
Type
Description & Constraints
Observation
0..*
USCoreLaboratoryResultObservationProfile
Measurements and simple assertions
meta
security
1..*
(Slice Definition)
Security Labels applied to this resource
Slice: Unordered, Open by pattern:system
security:TransplantCenter
S
1..1
Coding
Security Labels applied to this resource
system
1..1
uri
Identity of the terminology system
Required Pattern: http://terminology.cibmtr.org/codesystem/transplant-center
code
I
1..1
code
Symbol in syntax defined by the system
sec-rc: Use transplant center identifier for security tag
Documentation for this format
Other representations of profile: CSV, Excel, Schematron
3.2.1.1.1 Terminology Bindings
Path
Conformance
ValueSet / Code
Observation.meta.security
extensible
All Security Labels
Observation.meta.security:TransplantCenter
extensible
All Security Labels
Observation.meta.tag
example
CommonTags
Observation.language
preferred
CommonLanguages
Max Binding: AllLanguages
Observation.status
required
ObservationStatus
Observation.category
preferred
ObservationCategoryCodes


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fhir.nmdp.org/ig/cibmtr-reporting/index.html
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The CIBMTR facilitates critical observational and interventional research through scientific and statistical expertise, a large network of transplant centers, and a unique and extensive clinical outcomes database. CIBMTR collects data for HCT recipients using a web-based manual data entry system (FormsNet) and an automated messaging system (A Growable Network Information System (AGNIS)).
CIBMTR is now collecting data through the HL7 Fast Healthcare Interoperability Resources® (FHIR®) standard. This IG describes the technical specification that CIBMTR accepts from third party clients producing FHIR resources for reporting HCT data. It is primarily based on the US Core FHIR IG which defines the minimum conformance requirements for accessing patient data.
Initially, this IG is focused on information required for sending Patient demographic data that is necessary for creating or retrieving a CIBMTR Research ID (CRID), and for reporting Patient resources with a known CRID. We have also begun specifying a profile for reporting laboratory observations, focusing on a set of variables that have been identified as a priority for the CIBMTR Data Transformation Initiative in Fiscal Year 2021.
1.2 CRID Lookup/Creation
A REST API is available for enabling users to lookup and/or create CRIDs based on patient demographic information. This is described in Direct FHIR for STU3 Guide. This API does not use FHIR Patient resources, but instead uses a special JSON structure. We are working on true FHIR interface for this function in the future where Patient FHIR resources may be submitted. In the meantime, please use the method described here.
1.3 Highlighted Artifacts
1.3.1 Profiles
CIBMTR Patient
CIBMTR Observation Laboratory Results
CIBMTR Observation for Priority Variables
1.3.2 ValueSets
CIBMTR Priority Variables for FY21
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fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-obs-priority-variables.html
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This page is part of the CIMBTRReporting (v0.1.4: Trial Use 1) based on FHIR R4. This is the current published version. For a full list of available versions, see the Directory of published versions
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3.1.1 Resource Profile: CIBMTR Observation Laboratory Results: Priority Variables
Defining URL: http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables
Version: 0.1.4
Name: CIBMTRObsPriorityVariables
Title: CIBMTR Observation Laboratory Results: Priority Variables
Status: Active as of 2021-06-25T13:41:27-05:00
Definition: CIBMTR Observation Laboratory Results: Priority Variables
Publisher: The Medical College of Wisconsin, Inc. and the National Marrow Donor Program
Source Resource: XML / JSON / Turtle
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables
3.1.1.1 Formal Views of Profile Content
Description of Profiles, Differentials, Snapshots and how the different presentations work.
Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
This structure is derived from CIBMTRObservationLab
Name
Flags
Card.
Type
Description & Constraints
Observation
0..*
CIBMTRObservationLab
Measurements and simple assertions
code
coding
0..*
Coding
Code defined by a terminology system


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fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-obs-priority-variables.html
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables
3.1.1.1 Formal Views of Profile Content
Description of Profiles, Differentials, Snapshots and how the different presentations work.
Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
This structure is derived from CIBMTRObservationLab
Name
Flags
Card.
Type
Description & Constraints
Observation
0..*
CIBMTRObservationLab
Measurements and simple assertions
code
coding
0..*
Coding
Code defined by a terminology system
Binding: CIBMTR Priority Variables Value Set (FY21)
(extensible)
Documentation for this format
Other representations of profile: CSV, Excel, Schematron
3.1.1.1.1 Terminology Bindings
Path
Conformance
ValueSet / Code
Observation.meta.security
extensible
All Security Labels
Observation.meta.security:TransplantCenter
extensible
All Security Labels
Observation.meta.tag
example
CommonTags
Observation.language
preferred
CommonLanguages
Max Binding: AllLanguages


CIBMTR-REPORTING\CIBMTR
fhir.nmdp.org/ig/cibmtr-reporting/ValueSet-cibmtr-priority-variables-2021.html
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This page is part of the CIMBTRReporting (v0.1.4: Trial Use 1) based on FHIR R4. This is the current published version. For a full list of available versions, see the Directory of published versions
Narrative Content
XML
JSON
TTL
3.16.1 ValueSet: CIBMTR Priority Variables Value Set (FY21)
Summary
Defining URL: http://fhir.nmdp.org/ig/cibmtr-reporting/ValueSet/cibmtr-priority-variables-2021
Version: 0.1.4
Name: PriorityVariables2021
Title: CIBMTR Priority Variables Value Set (FY21)
Status: Active as of 2021-06-25T13:41:27-05:00
Definition: Priority Variables for CIBMTR (FY21)
Publisher: The Medical College of Wisconsin, Inc. and the National Marrow Donor Program
Source Resource: XML / JSON / Turtle
References
CIBMTR Observation Laboratory Results: Priority Variables
3.16.1.1 Introduction
CIBMTR currently supports submission of lab measurements collected prior to and post-HCT transfusion. When submitting FHIR Observation resources, one of the below supported LOINC codes must be used in the code section of the resource. Selecting the correct LOINC code to use to represent the clinical concept of the lab data should be done by someone clinically trained to understand the lab measurement and corresponding LOINC code. Lab quantities should always include the corresponding unit of measure coded using the UCUM standard vocabulary.



3.16.1.1 Introduction

CIBMTR currently supports submission of lab measurements collected prior to and post-HCT transfusion. When submitting FHIR Observation resources, one of the below supported LOINC codes must be used in the code section of the resource. Selecting the correct LOINC code to use to represent the clinical concept of the lab data should be done by someone clinically trained to understand the lab measurement and corresponding LOINC code. Lab quantities should always include the corresponding unit of measure coded using the UCUM standard vocabulary.

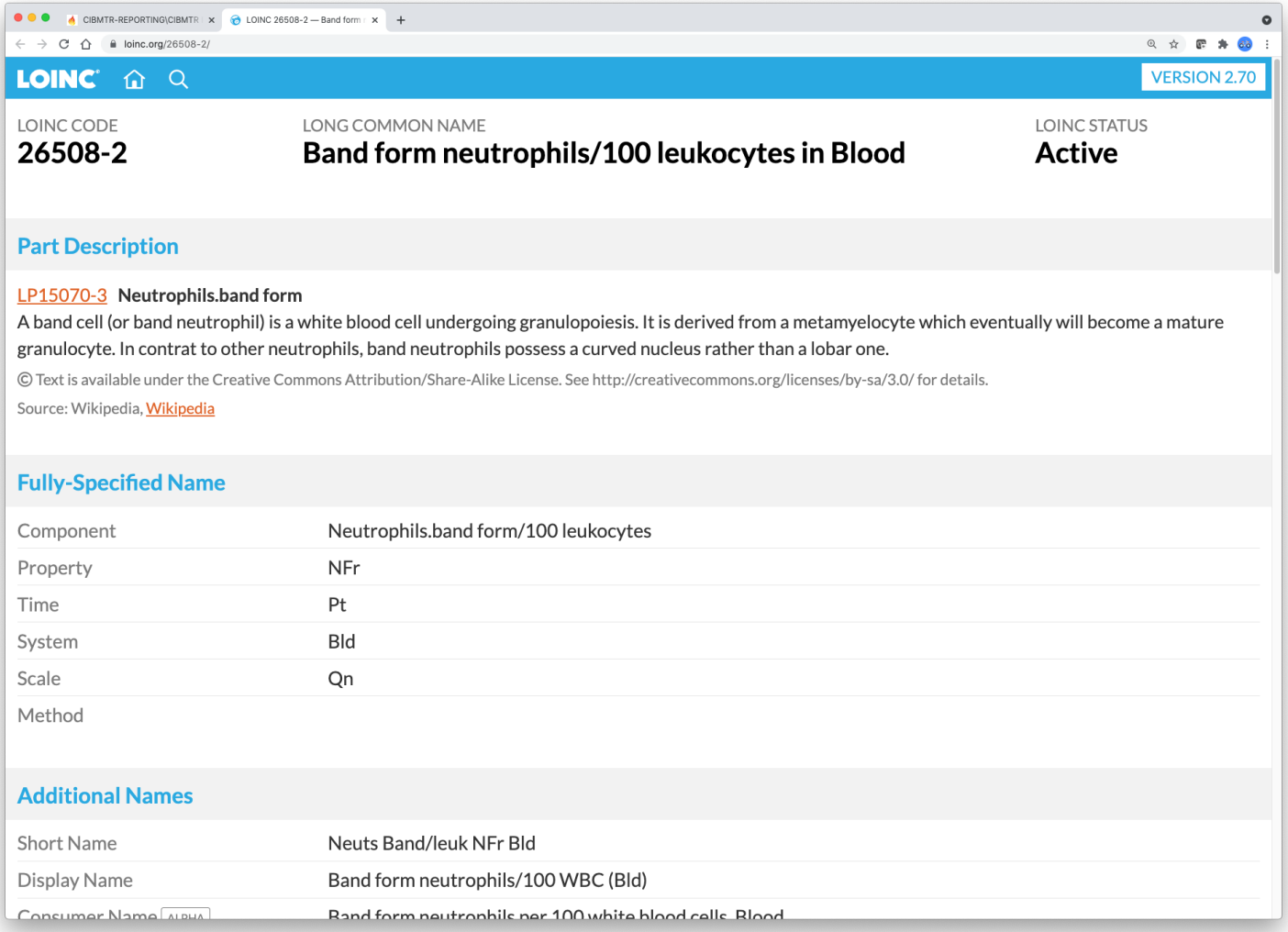
The codes described here represent variables that have been identified as a priority for the CIBMTR Data Transformation Initiative in Fiscal Year 2021. They include codes from LOINC panels describing Complete Blood Count (CBC) (see below) and other codes representing similar concepts but are not found in these panels.

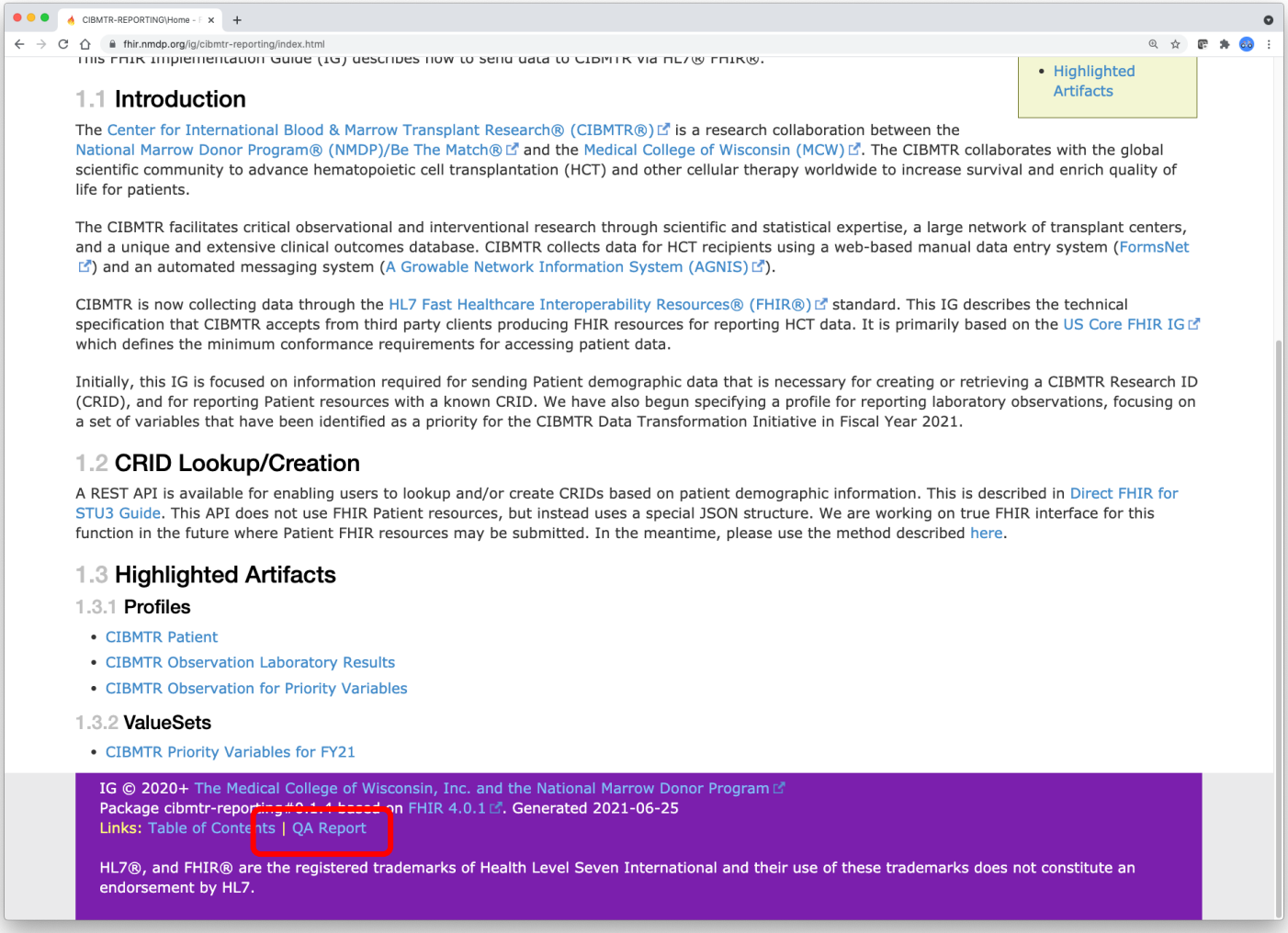
Panel Code	Name
57782-5	CBC W Ordered Manual Differential panel – Blood
57021-8	CBC W Auto Differential panel – Blood
69742-5	CBC W Differential panel, method unspecified – Blood

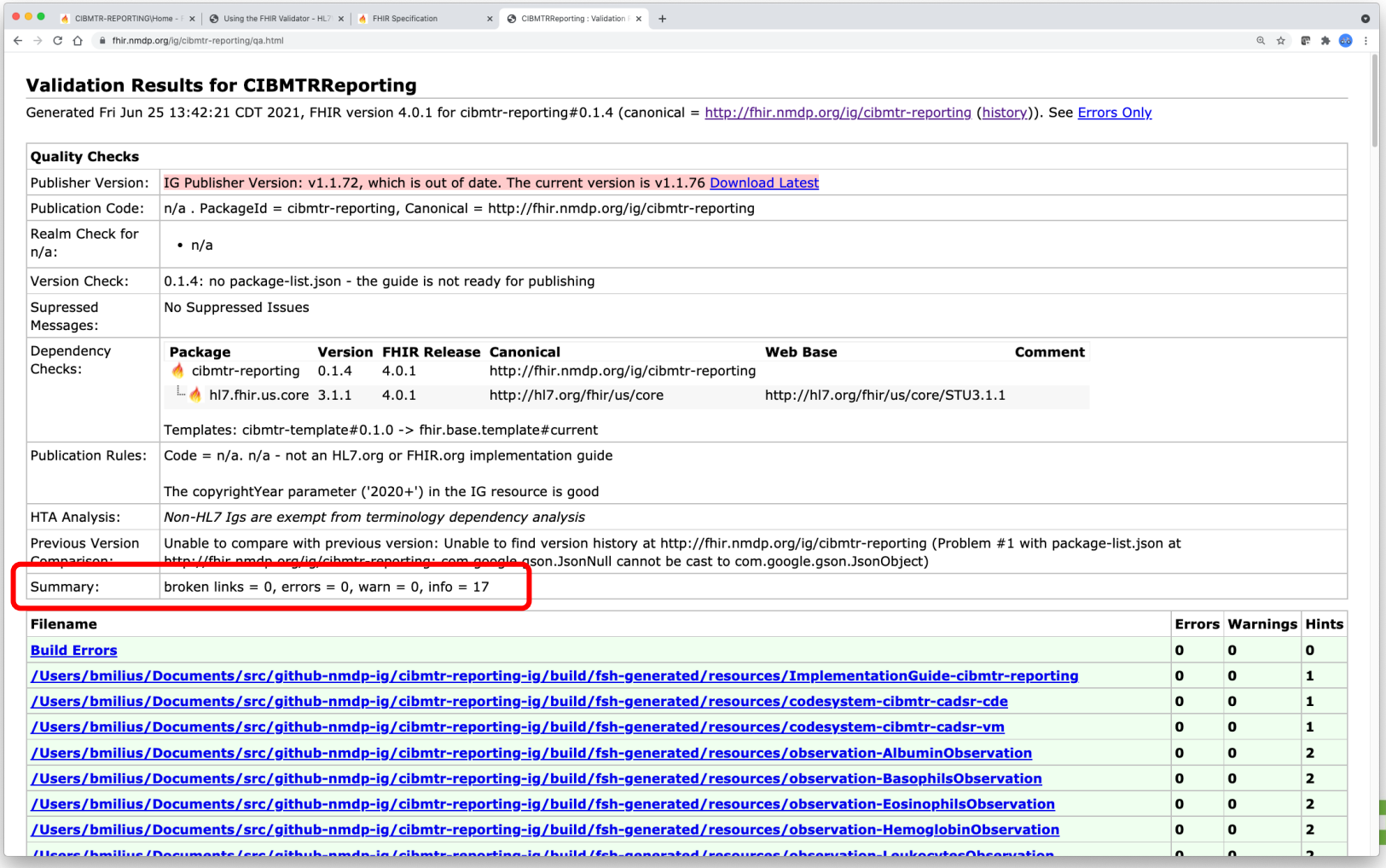
3.16.1.2 Logical Definition (CLD)

- Include these codes as defined in `http://loinc.org`

Code	Display
26508-2	Band form neutrophils/100 leukocytes in Blood
35332-6	Band form neutrophils/100 leukocytes in Blood by Automated count
764-1	Band form neutrophils/100 leukocytes in Blood by Manual count
30180-4	Basophils/100 leukocytes in Blood
706-2	Basophils/100 leukocytes in Blood by Automated count
707-0	Basophils/100 leukocytes in Blood by Manual count
26446-5	Blasts/100 leukocytes in Blood
709-6	Blasts/100 leukocytes in Blood by Manual count
26449-9	Eosinophils [#/volume] in Blood
711-2	Eosinophils [#/volume] in Blood by Automated count
712-0	Eosinophils [#/volume] in Blood by Manual count
26450-7	Eosinophils/100 leukocytes in Blood
713-8	Eosinophils/100 leukocytes in Blood by Automated count
714-6	Eosinophils/100 leukocytes in Blood by Manual count
4576-5	Hemoglobin F/Hemoglobin.total in Blood
71865-0	Hemoglobin F/Hemoglobin.total [Pure mass fraction] in Blood by Electrophoresis


CIBMTR-REPORTING\CIBMTR
LOINC 26508-2 — Band form
loinc.org/26508-2/
LOINC
VERSION 2.70
LOINC CODE
26508-2
LONG COMMON NAME
Band form neutrophils/100 leukocytes in Blood
LOINC STATUS
Active
Part Description
LP15070-3 Neutrophils.band form
A band cell (or band neutrophil) is a white blood cell undergoing granulopoiesis. It is derived from a metamyelocyte which eventually will become a mature granulocyte. In contrat to other neutrophils, band neutrophils possess a curved nucleus rather than a lobar one.
© Text is available under the Creative Commons Attribution/Share-Alike License. See http://creativecommons.org/licenses/by-sa/3.0/ for details.
Source: Wikipedia, Wikipedia
Fully-Specified Name
Component Neutrophils.band form/100 leukocytes
Property NFr
Time Pt
System Bld
Scale Qn
Method
Additional Names
Short Name Neuts Band/leuk NFr Bld
Display Name Band form neutrophils/100 WBC (Bld)
Consumer Name ALPHA Band form neutrophils per 100 white blood cells, Blood


CIBMTR-REPORTING\Home -
fhir.nmdp.org/ig/cibmtr-reporting/index.html
This FHIR Implementation Guide (IG) describes how to send data to CIBMTR via HL7® FHIR®.
Highlighted Artifacts
1.1 Introduction
The Center for International Blood & Marrow Transplant Research® (CIBMTR®) is a research collaboration between the National Marrow Donor Program® (NMDP)/Be The Match® and the Medical College of Wisconsin (MCW). The CIBMTR collaborates with the global scientific community to advance hematopoietic cell transplantation (HCT) and other cellular therapy worldwide to increase survival and enrich quality of life for patients.
The CIBMTR facilitates critical observational and interventional research through scientific and statistical expertise, a large network of transplant centers, and a unique and extensive clinical outcomes database. CIBMTR collects data for HCT recipients using a web-based manual data entry system (FormsNet) and an automated messaging system (A Growable Network Information System (AGNIS)).
CIBMTR is now collecting data through the HL7 Fast Healthcare Interoperability Resources® (FHIR®) standard. This IG describes the technical specification that CIBMTR accepts from third party clients producing FHIR resources for reporting HCT data. It is primarily based on the US Core FHIR IG which defines the minimum conformance requirements for accessing patient data.
Initially, this IG is focused on information required for sending Patient demographic data that is necessary for creating or retrieving a CIBMTR Research ID (CRID), and for reporting Patient resources with a known CRID. We have also begun specifying a profile for reporting laboratory observations, focusing on a set of variables that have been identified as a priority for the CIBMTR Data Transformation Initiative in Fiscal Year 2021.
1.2 CRID Lookup/Creation
A REST API is available for enabling users to lookup and/or create CRIDs based on patient demographic information. This is described in Direct FHIR for STU3 Guide. This API does not use FHIR Patient resources, but instead uses a special JSON structure. We are working on true FHIR interface for this function in the future where Patient FHIR resources may be submitted. In the meantime, please use the method described here.
1.3 Highlighted Artifacts
1.3.1 Profiles
CIBMTR Patient
CIBMTR Observation Laboratory Results
CIBMTR Observation for Priority Variables
1.3.2 ValueSets
CIBMTR Priority Variables for FY21
IG © 2020+ The Medical College of Wisconsin, Inc. and the National Marrow Donor Program
Package cibmtr-reporting#0.1.1 based on FHIR 4.0.1. Generated 2021-06-25
Links: Table of Contents | QA Report
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Validation Results for CIBMTRReporting

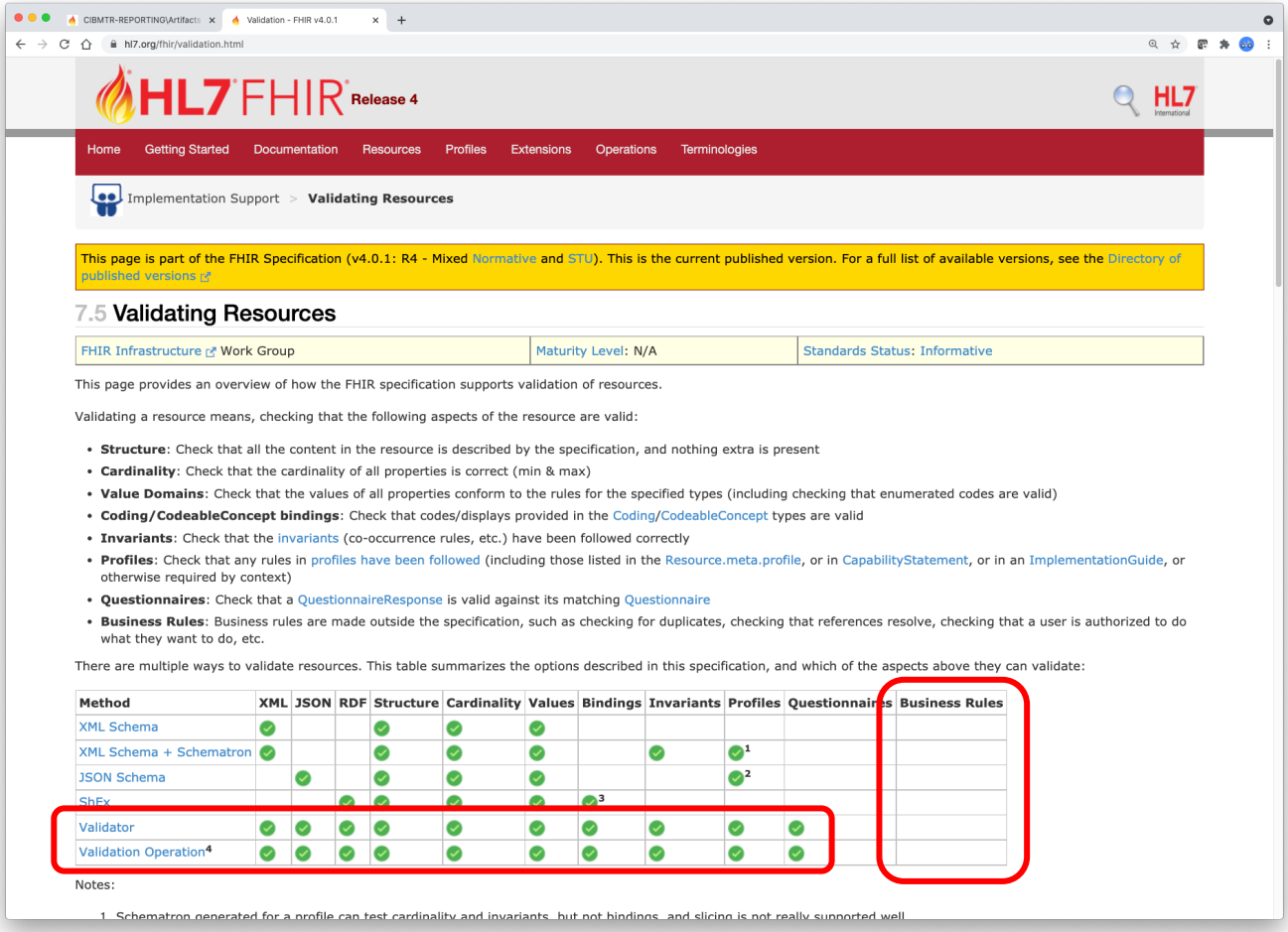
Generated Fri Jun 25 13:42:21 CDT 2021, FHIR version 4.0.1 for cibmtr-reporting#0.1.4 (canonical = http://fhir.nmdp.org/ig/cibmtr-reporting (history)). See Errors Only

Quality Checks	
Publisher Version:	IG Publisher Version: v1.1.72, which is out of date. The current version is v1.1.76 Download Latest
Publication Code:	n/a . PackageId = cibmtr-reporting, Canonical = http://fhir.nmdp.org/ig/cibmtr-reporting
Realm Check for n/a:	• n/a
Version Check:	0.1.4: no package-list.json - the guide is not ready for publishing
Supressed Messages:	No Suppressed Issues
Dependency Checks:	Package / Version / FHIR Release / Canonical / Web Base / Comment cibmtr-reporting 0.1.4 4.0.1 http://fhir.nmdp.org/ig/cibmtr-reporting hl7.fhir.us.core 3.1.1 4.0.1 http://hl7.org/fhir/us/core http://hl7.org/fhir/us/core/STU3.1.1 Templates: cibmtr-template#0.1.0 -> fhir.base.template#current
Publication Rules:	Code = n/a. n/a - not an HL7.org or FHIR.org implementation guide The copyrightYear parameter ('2020+') in the IG resource is good
HTA Analysis:	*Non-HL7 Igs are exempt from terminology dependency analysis*
Previous Version Comparison:	Unable to compare with previous version: Unable to find version history at http://fhir.nmdp.org/ig/cibmtr-reporting (Problem #1 with package-list.json at http://fhir.nmdp.org/ig/cibmtr-reporting: com.google.gson.JsonNull cannot be cast to com.google.gson.JsonObject)
Summary:	broken links = 0, errors = 0, warn = 0, info = 17

Filename	Errors	Warnings	Hints
Build Errors	0	0	0
/Users/bmilius/Documents/src/github-nmdp-ig/cibmtr-reporting-ig/build/fsh-generated/resources/ImplementationGuide-cibmtr-reporting	0	0	1
/Users/bmilius/Documents/src/github-nmdp-ig/cibmtr-reporting-ig/build/fsh-generated/resources/codesystem-cibmtr-cadsr-cde	0	0	1
/Users/bmilius/Documents/src/github-nmdp-ig/cibmtr-reporting-ig/build/fsh-generated/resources/codesystem-cibmtr-cadsr-vm	0	0	1
/Users/bmilius/Documents/src/github-nmdp-ig/cibmtr-reporting-ig/build/fsh-generated/resources/observation-AlbuminObservation	0	0	2
/Users/bmilius/Documents/src/github-nmdp-ig/cibmtr-reporting-ig/build/fsh-generated/resources/observation-BasophilsObservation	0	0	2
/Users/bmilius/Documents/src/github-nmdp-ig/cibmtr-reporting-ig/build/fsh-generated/resources/observation-EosinophilsObservation	0	0	2
/Users/bmilius/Documents/src/github-nmdp-ig/cibmtr-reporting-ig/build/fsh-generated/resources/observation-HemoglobinObservation	0	0	2



FHIR Validation



Validation - FHIR v4.0.1

hl7.org/fhir/validation.html

HL7 FHIR Release 4

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Implementation Support > **Validating Resources**

This page is part of the FHIR Specification (v4.0.1: R4 - Mixed Normative and STU). This is the current published version. For a full list of available versions, see the Directory of published versions

7.5 Validating Resources

FHIR Infrastructure Work Group	Maturity Level: N/A	Standards Status: Informative

This page provides an overview of how the FHIR specification supports validation of resources.

Validating a resource means, checking that the following aspects of the resource are valid:

- **Structure**: Check that all the content in the resource is described by the specification, and nothing extra is present
- **Cardinality**: Check that the cardinality of all properties is correct (min & max)
- **Value Domains**: Check that the values of all properties conform to the rules for the specified types (including checking that enumerated codes are valid)
- **Coding/CodeableConcept bindings**: Check that codes/displays provided in the Coding/CodeableConcept types are valid
- **Invariants**: Check that the invariants (co-occurrence rules, etc.) have been followed correctly
- **Profiles**: Check that any rules in profiles have been followed (including those listed in the Resource.meta.profile, or in CapabilityStatement, or in an ImplementationGuide, or otherwise required by context)
- **Questionnaires**: Check that a QuestionnaireResponse is valid against its matching Questionnaire
- **Business Rules**: Business rules are made outside the specification, such as checking for duplicates, checking that references resolve, checking that a user is authorized to do what they want to do, etc.

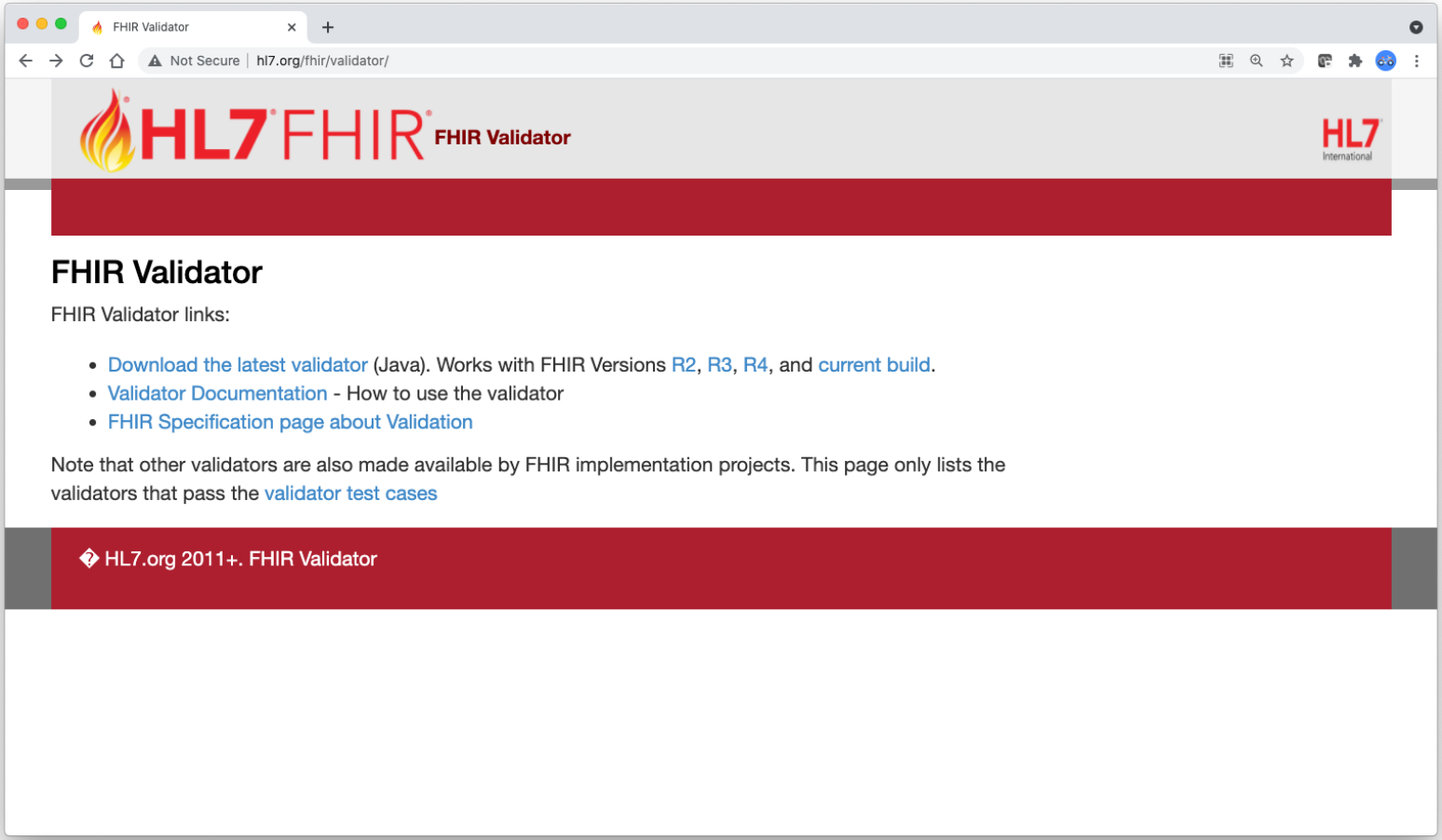
There are multiple ways to validate resources. This table summarizes the options described in this specification, and which of the aspects above they can validate:

Method	XML	JSON	RDF	Structure	Cardinality	Values	Bindings	Invariants	Profiles	Questionnaires	Business Rules
XML Schema	✓			✓	✓	✓					
XML Schema + Schematron	✓			✓	✓	✓		✓	✓[1]		
JSON Schema		✓		✓	✓	✓			✓[2]		
ShEx			✓	✓	✓	✓	✓[3]				
Validator	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Validation Operation[4]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	

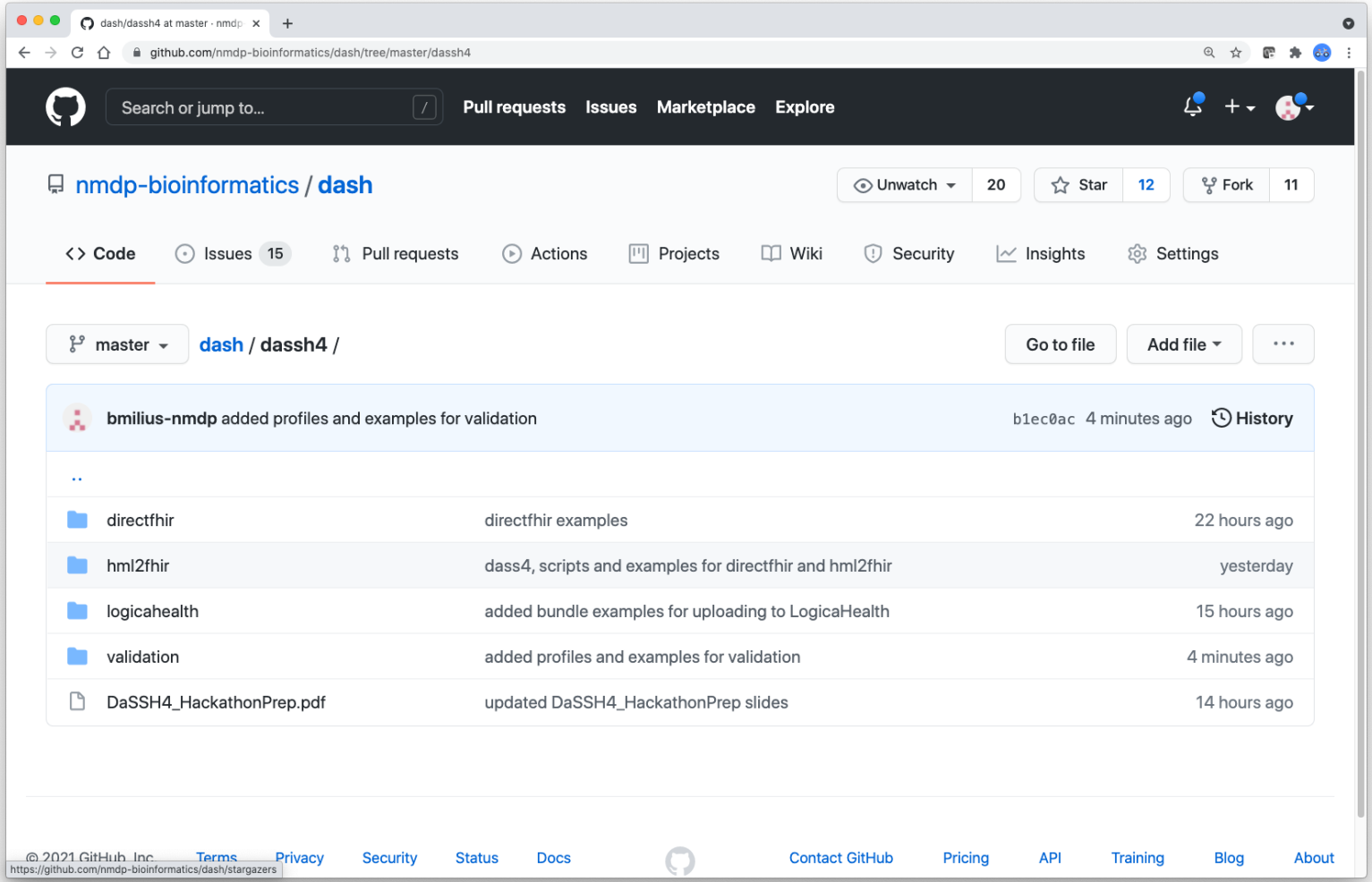
Notes:

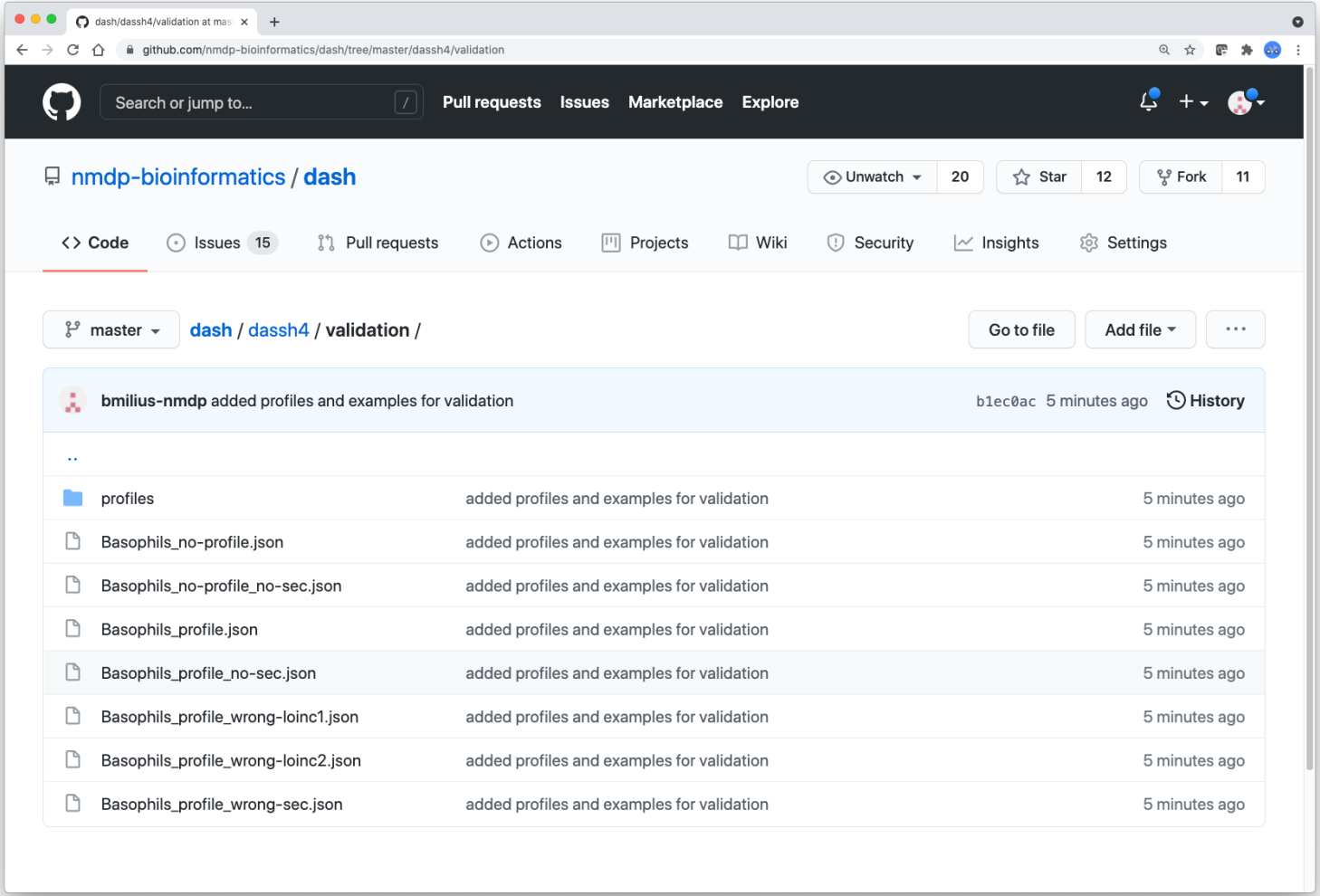
1. Schematron generated for a profile can test cardinality and invariants, but not bindings, and slicing is not really supported well

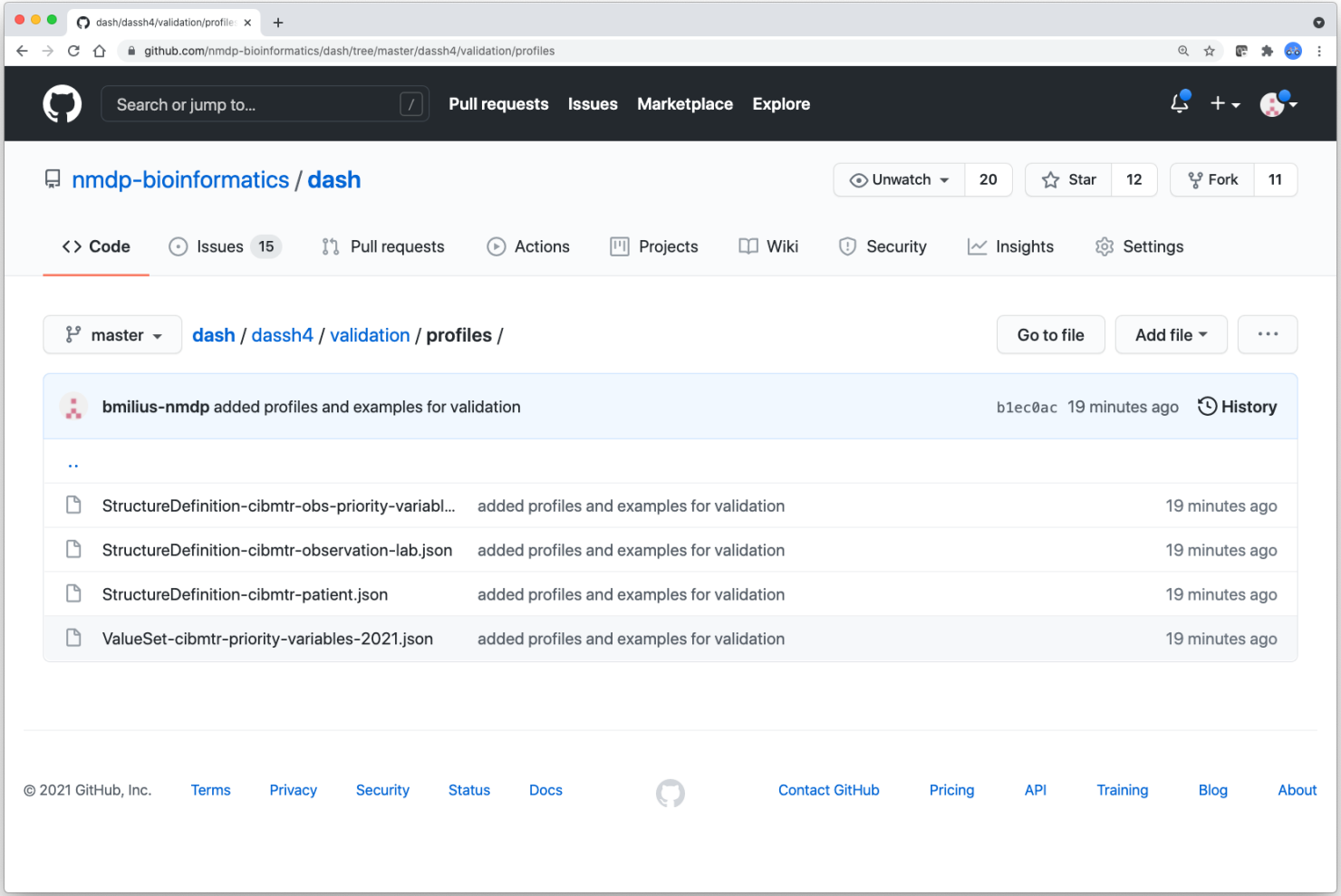
http://hl7.org/fhir/validation.html

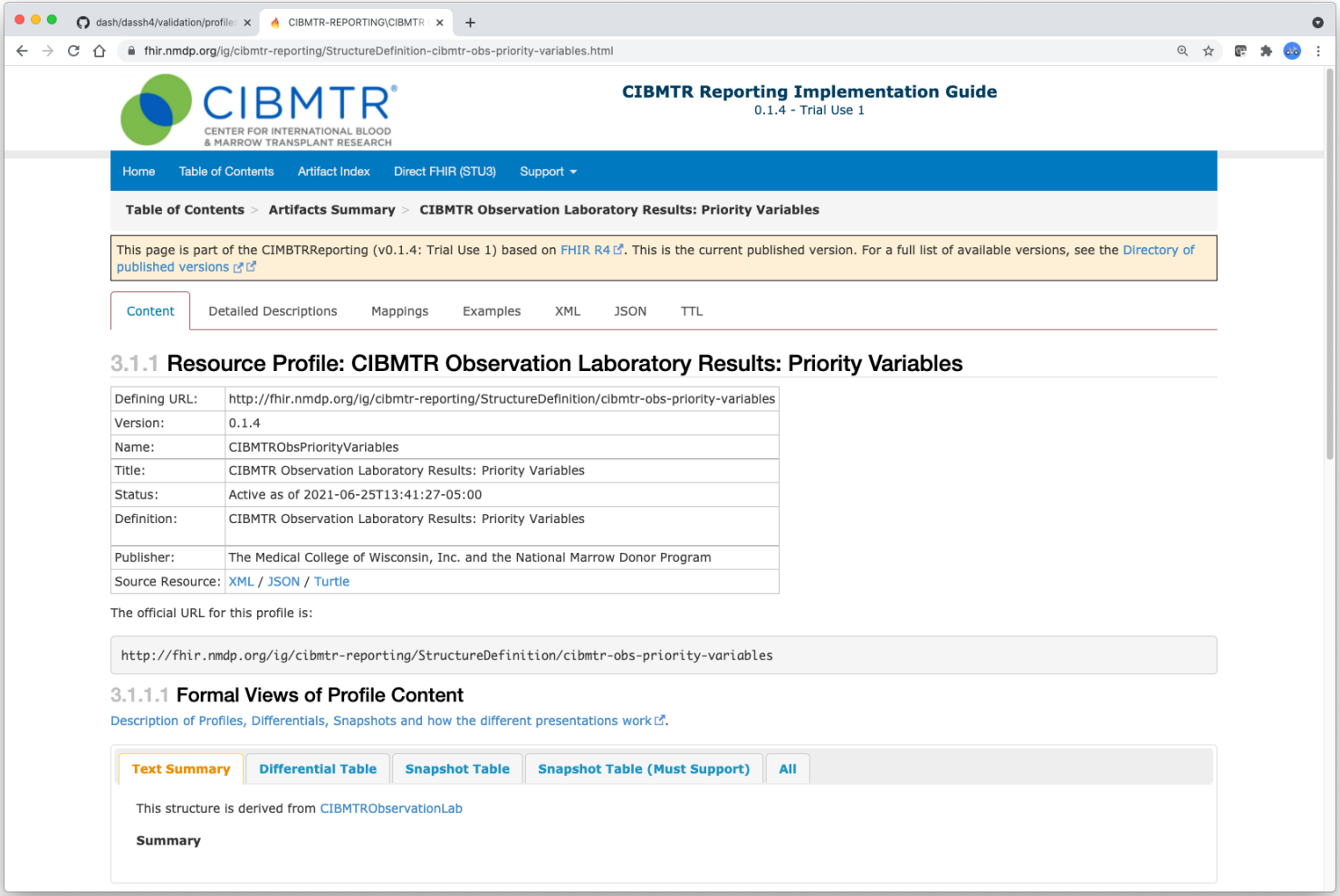


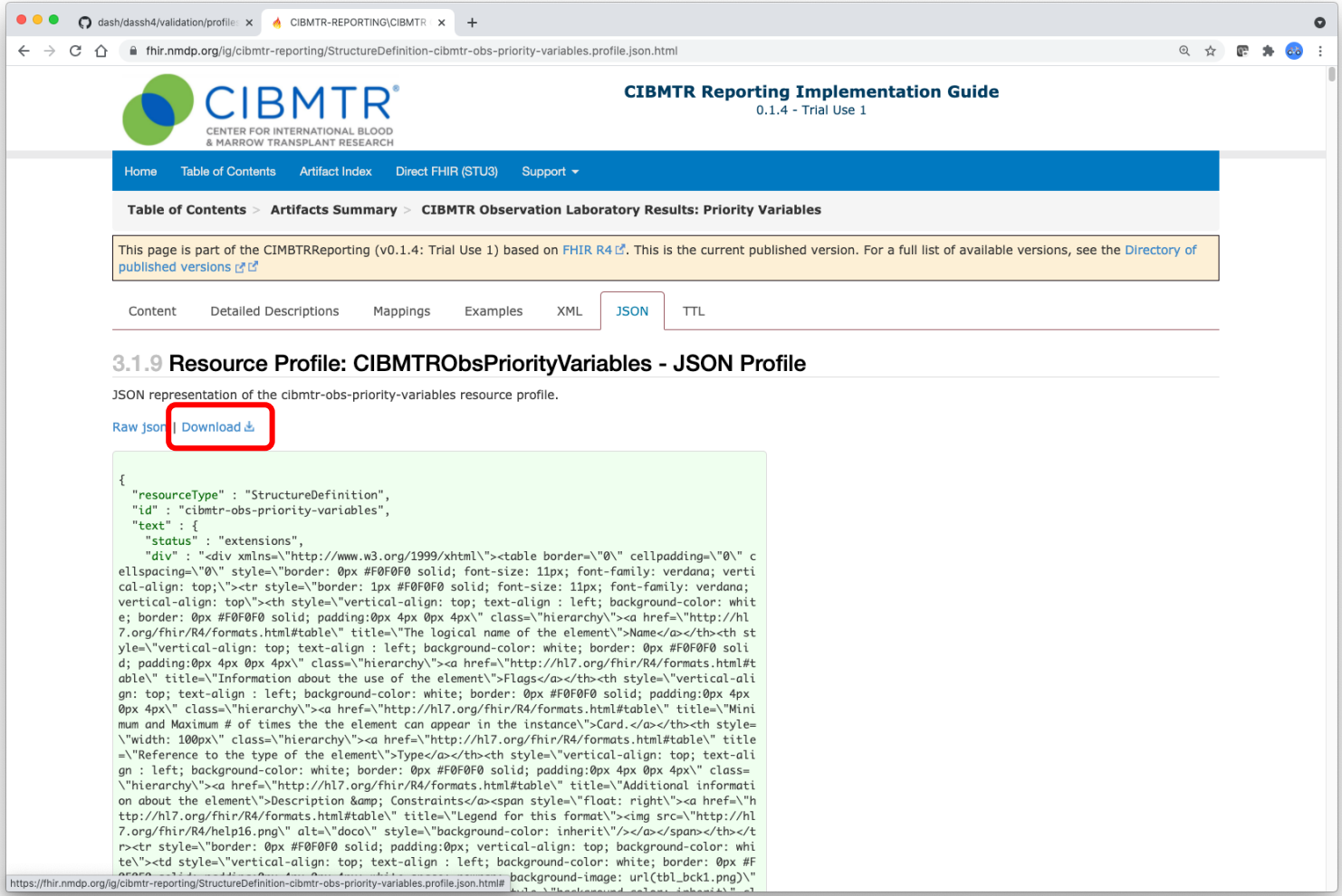
http://hl7.org/fhir/validator/

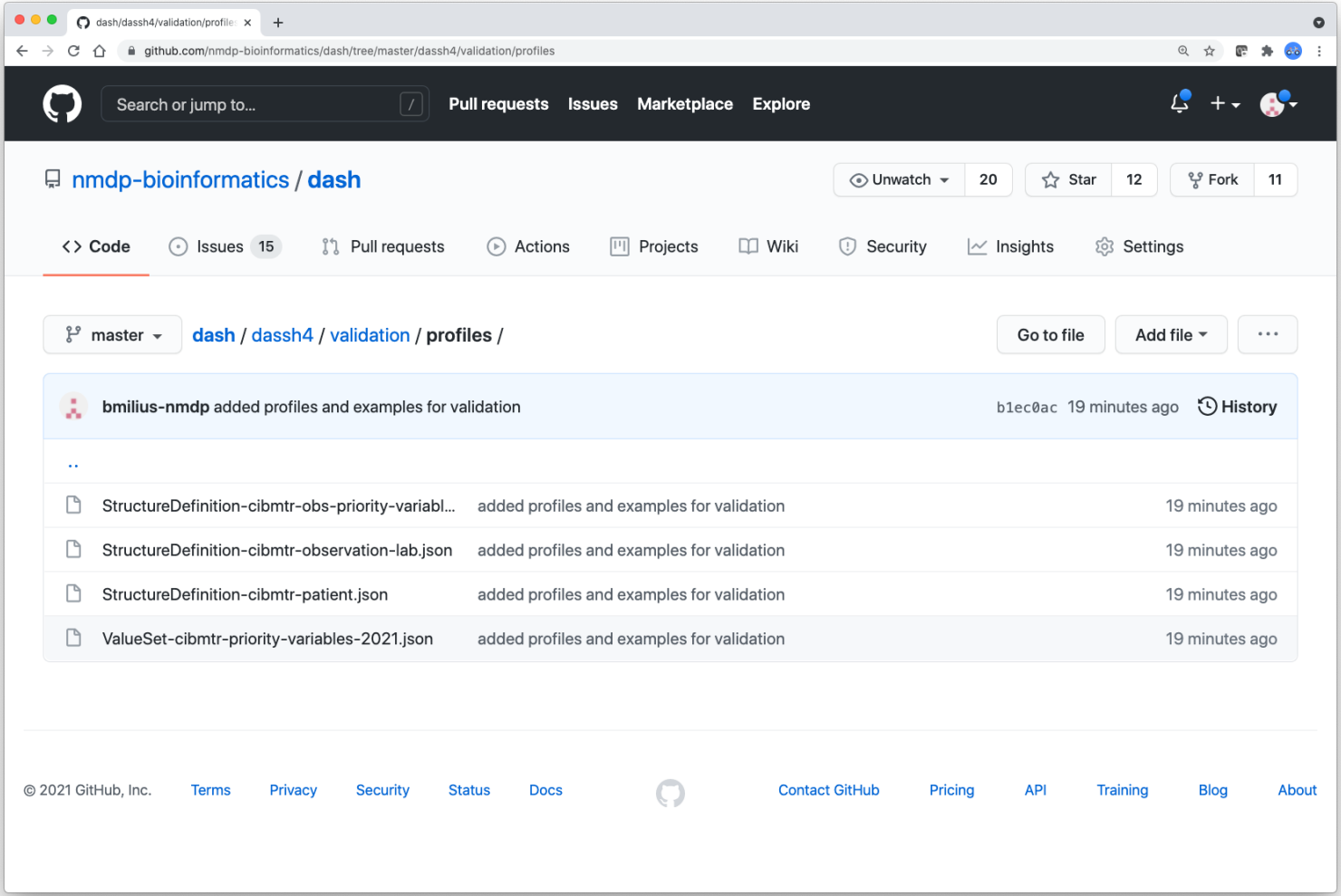

dash/dassh4 at master · nmdp
github.com/nmdp-bioinformatics/dash/tree/master/dassh4
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hml2fhir
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yesterday
logicahealth
added bundle examples for uploading to LogicaHealth
15 hours ago
validation
added profiles and examples for validation
4 minutes ago
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updated DaSSH4_HackathonPrep slides
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Basophils_no-profile_no-sec.json added profiles and examples for validation 5 minutes ago
Basophils_profile.json added profiles and examples for validation 5 minutes ago
Basophils_profile_no-sec.json added profiles and examples for validation 5 minutes ago
Basophils_profile_wrong-loinc1.json added profiles and examples for validation 5 minutes ago
Basophils_profile_wrong-loinc2.json added profiles and examples for validation 5 minutes ago
Basophils_profile_wrong-sec.json added profiles and examples for validation 5 minutes ago


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StructureDefinition-cibmtr-observation-lab.json added profiles and examples for validation 19 minutes ago
StructureDefinition-cibmtr-patient.json added profiles and examples for validation 19 minutes ago
ValueSet-cibmtr-priority-variables-2021.json added profiles and examples for validation 19 minutes ago
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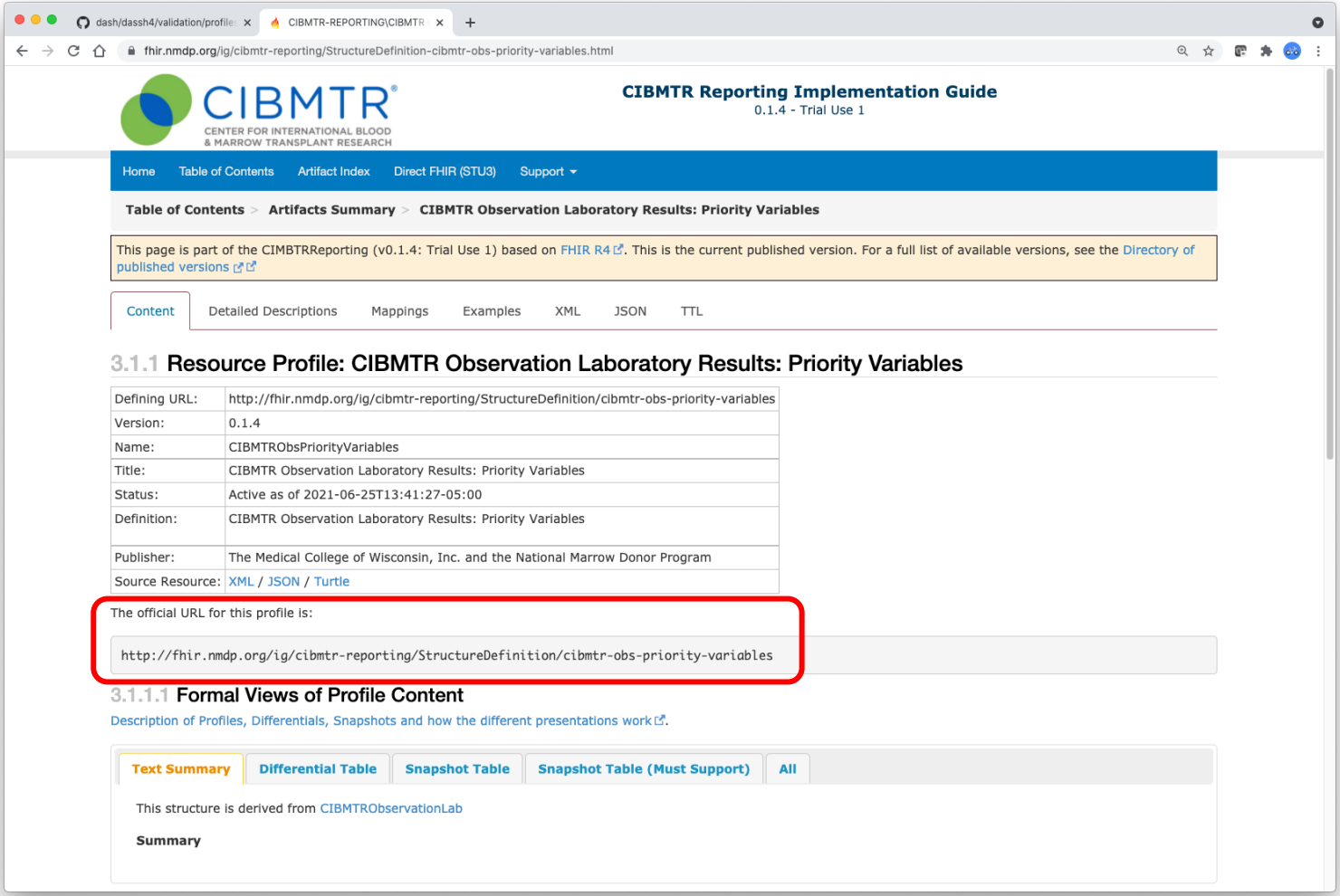

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Version: 0.1.4
Name: CIBMTRObsPriorityVariables
Title: CIBMTR Observation Laboratory Results: Priority Variables
Status: Active as of 2021-06-25T13:41:27-05:00
Definition: CIBMTR Observation Laboratory Results: Priority Variables
Publisher: The Medical College of Wisconsin, Inc. and the National Marrow Donor Program
Source Resource: XML / JSON / Turtle
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables
3.1.1.1 Formal Views of Profile Content
Description of Profiles, Differentials, Snapshots and how the different presentations work.
Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
This structure is derived from CIBMTRObservationLab
Summary


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fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-obs-priority-variables.profile.json.html
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3.1.9 Resource Profile: CIBMTRObsPriorityVariables - JSON Profile
JSON representation of the cibmtr-obs-priority-variables resource profile.
Raw Json | Download
{
"resourceType" : "StructureDefinition",
"id" : "cibmtr-obs-priority-variables",
"text" : {
"status" : "extensions",
"div" : "<div xmlns=\"http://www.w3.org/1999/xhtml\"><table border=\"0\" cellpadding=\"0\" cellspacing=\"0\" style=\"border: 0px #F0F0F0 solid; font-size: 11px; font-family: verdana; vertical-align: top;\"><tr style=\"border: 1px #F0F0F0 solid; font-size: 11px; font-family: verdana; vertical-align: top\"><th style=\"vertical-align: top; text-align : left; background-color: white; border: 0px #F0F0F0 solid; padding:0px 4px 0px 4px\" class=\"hierarchy\">Name</th><th style=\"vertical-align: top; text-align : left; background-color: white; border: 0px #F0F0F0 solid; padding:0px 4px 0px 4px\" class=\"hierarchy\">Flags</th><th style=\"vertical-align: top; text-align : left; background-color: white; border: 0px #F0F0F0 solid; padding:0px 4px 0px 4px\" class=\"hierarchy\">Card.</th><th style=\"width: 100px\" class=\"hierarchy\">Type</th><th style=\"vertical-align: top; text-align : left; background-color: white; border: 0px #F0F0F0 solid; padding:0px 4px 0px 4px\" class=\"hierarchy\">Description & Constraints<span style=\"float: right\"></th></tr><tr style=\"border: 0px #F0F0F0 solid; padding:0px; vertical-align: top; background-color: white\"><td style=\"vertical-align: top; text-align : left; background-color: white; border: 0px #F
background-image: url(tbl_bck1.png)\"
https://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-obs-priority-variables.profile.json.html#


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ValueSet-cibmtr-priority-variables-2021.json added profiles and examples for validation 19 minutes ago
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```
{
    "resourceType": "Observation",
    "meta": {
        "profile": ["http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables"],
        "security": [
            {
                "system": "http://terminology.cibmtr.org/codesystem/transplant-center",
                "code": "rc_12002"
            }
        ]
    },
    "status": "final",
    "category": [
        {"coding": [
            {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "laboratory"
            }
        ]}
    ],
    "code": {"coding": [
        {
            "system": "http://loinc.org",
            "code": "30180-4",
            "display": "Basophils/100 leukocytes in Blood"
        }
    ]},
    "subject": {"reference": "Patient/123"},
    "effectiveDateTime": "2020-07-21T14:17:00Z",
    "valueQuantity": {
        "value": 1,
        "unit": "percent",
        "system": "http://unitsofmeasure.org",
        "code": "%"
    }
}
```

Basophils_profile.json


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Defining URL: http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables
Version: 0.1.4
Name: CIBMTRObsPriorityVariables
Title: CIBMTR Observation Laboratory Results: Priority Variables
Status: Active as of 2021-06-25T13:41:27-05:00
Definition: CIBMTR Observation Laboratory Results: Priority Variables
Publisher: The Medical College of Wisconsin, Inc. and the National Marrow Donor Program
Source Resource: XML / JSON / Turtle
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables
3.1.1.1 Formal Views of Profile Content
Description of Profiles, Differentials, Snapshots and how the different presentations work.
Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
This structure is derived from CIBMTRObservationLab
Summary

```
$ java -jar validator_cli.jar Basophils_profile.json -version 4.0.1 -ig ./profiles

FHIR Validation tool Version 5.4.9 (Git# 296ab05f10f2). Built 2021-07-15T16:29:32.431Z (21 days old)
  Java:   1.8.0_172 from /Library/Java/JavaVirtualMachines/jdk1.8.0_172.jdk/Contents/Home/jre on x86_64
(64bit). 3641MB available
  ...
  Params: Basophils_profile.json -version 4.0.1 -ig ./profiles
Loading
  Load FHIR v4.0 from hl7.fhir.r4.core#4.0.1 - 4575 resources (00:04.0142)
  Load hl7.terminology#2.1.0 - 3767 resources (00:00.0997)
  Terminology server http://tx.fhir.org - Version 1.9.378 (00:01.0333)
  Load ./profiles - 4 resources (00:00.0117)
  Get set...  go (00:00.0032)
Validating
  Validate Basophils_profile.json 00:01.0203
Done. Times: Loading: 00:06.0825, validation: 00:01.0204. Memory = 339Mb

Success: 0 errors, 1 warnings, 1 notes
  Information @ Observation.meta.security[0] (line 8, col10) : Code System URI
'http://terminology.cibmtr.org/codesystem/transplant-center' is unknown so the code cannot be validated
  Warning @ Observation.meta.security[0] (line 8, col10) : The Coding provided
(http://terminology.cibmtr.org/codesystem/transplant-center#rc_12002) is not in the value set
http://hl7.org/fhir/ValueSet/security-labels, and a code should come from this value set unless it has
no suitable code (note that the validator cannot judge what is suitable).  (error message = Not in value
set http://hl7.org/fhir/ValueSet/security-labels)
```



```
{
    "resourceType": "Observation",
    "meta": {
        "profile": ["http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables"]
     },
    "status": "final",
    "category": [
        {"coding": [
            {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "laboratory"
            }
        ]}
    ],
    "code": {"coding": [
        {
            "system": "http://loinc.org",
            "code": "30180-4",
            "display": "Basophils/100 leukocytes in Blood"
        }
    ]},
    "subject": {"reference": "Patient/123"},
    "effectiveDateTime": "2020-07-21T14:17:00Z",
    "valueQuantity": {
        "value": 1,
        "unit": "percent",
        "system": "http://unitsofmeasure.org",
        "code": "%"
    }
}
```


Basophils_profile_no-sec.json



```
$ java –jar validator_cli.jar Basophils_profile_no-sec.json -version 4.0.1 -ig ./profiles

 FHIR Validation tool Version 5.4.9 (Git# 296ab05f10f2). Built 2021-07-15T16:29:32.431Z (21 days old)
  Java:   1.8.0_172 from /Library/Java/JavaVirtualMachines/jdk1.8.0_172.jdk/Contents/Home/jre on x86_64
(64bit). 3641MB available
  ...
  Params: Basophils_profile_no-sec.json -version 4.0.1 -ig ./profiles
Loading
  Load FHIR v4.0 from hl7.fhir.r4.core#4.0.1 - 4575 resources (00:04.0094)
  Load hl7.terminology#2.1.0 - 3767 resources (00:01.0167)
  Terminology server http://tx.fhir.org - Version 1.9.378 (00:01.0310)
  Load ./profiles - 4 resources (00:00.0131)
  Get set...  go (00:00.0039)
Validating
  Validate Basophils_profile_no-sec.json 00:01.0837
Done. Times: Loading: 00:06.0953, validation: 00:01.0838. Memory = 329Mb

*FAILURE*: 2 errors, 0 warnings, 0 notes
  Error @ Observation.meta (line 4, col19) : Observation.meta.security: minimum required = 1, but only found
0 (from http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables)
  Error @ Observation.meta (line 4, col19) : Observation.meta.security:TransplantCenter: minimum required =
1, but only found 0 (from http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-
variables)
```



```
{
    "resourceType": "Observation",
    "meta": {
        "profile": ["http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables"],
        "security": [
            {
                "system": "http://terminology.cibmtr.org/codesystem/transplant-center",
                "code": "rc_12002"
            }
        ]
    },
    "status": "final",
    "category": [
        {"coding": [
            {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "laboratory"
            }
        ]}
    ],
    "code": {"coding": [
        {
            "system": "http://loinc.org",
            "code": "30180-4",
            "display": "Basophils/100 leukocytes in Blood"
        }
    ]},
    "subject": {"reference": "Patient/123"},
    "effectiveDateTime": "2020-07-21T14:17:00Z",
    "valueQuantity": {
        "value": 1,
        "unit": "percent",
        "system": "http://unitsofmeasure.org",
        "code": "%"
    }
}
```




```
{
    "resourceType": "Observation",
    "meta": {
        "profile": ["http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables"],
        "security": [
            {
                "system": "http://terminology.cibmtr.org/codesystem/transplant-center",
                "code": "rc_12002"
            }
        ]
    },
    "status": "final",
    "category": [
        {"coding": [
            {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "laboratory"
            }
        ]}
    ],
    "code": {"coding": [
        {
            "system": "http://loinc.org",
            "code": "30180",
            "display": "Basophils/100 leukocytes in Blood"
        }
    ]},
    "subject": {"reference": "Patient/123"},
    "effectiveDateTime": "2020-07-21T14:17:00Z",
    "valueQuantity": {
        "value": 1,
        "unit": "percent",
        "system": "http://unitsofmeasure.org",
        "code": "%"
    }
}
```


Basophils_profile_wrong-loinc1.json



```
$ java -jar validator_cli.jar Basophils_profile_wrong-loinc1.json -version 4.0.1 -ig ./profiles

FHIR Validation tool Version 5.4.9 (Git# 296ab05f10f2). Built 2021-07-15T16:29:32.431Z (21 days old)
  Java:   1.8.0_172 from /Library/Java/JavaVirtualMachines/jdk1.8.0_172.jdk/Contents/Home/jre on x86_64 (64bit). 3641MB available
  ...
  Params: Basophils_profile_wrong-loinc.json -version 4.0.1 -ig ./profiles
Loading
  Load FHIR v4.0 from hl7.fhir.r4.core#4.0.1 - 4575 resources (00:04.0225)
  Load hl7.terminology#2.1.0 - 3767 resources (00:01.0078)
  Terminology server http://tx.fhir.org - Version 1.9.378 (00:01.0286)
  Load ./profiles - 4 resources (00:00.0122)
  Get set...  go (00:00.0036)
Validating
  Validate Basophils_profile_wrong-loinc.json 00:01.0036
Done. Times: Loading: 00:06.0952, validation: 00:01.0036. Memory = 328Mb

*FAILURE*: 2 errors, 2 warnings, 1 notes
  Error @ Observation.code (line 26, col18) : The code "30180" is not valid in the system http://loinc.org; The code provided
(http://loinc.org#30180) is not valid in the value set 'LOINCCodes' (from http://tx.fhir.org/r4)
  Error @ Observation.code.coding[0] (line 27, col10) : The code "30180" is not valid in the system http://loinc.org; The code
provided (http://loinc.org#30180) is not valid in the value set 'All codes known to the system' (from http://tx.fhir.org/r4) for
'http://loinc.org#30180'
  Information @ Observation.meta.security[0] (line 8, col10) : Code System URI 'http://terminology.cibmtr.org/codesystem/transplant-
center' is unknown so the code cannot be validated
  Warning @ Observation.meta.security[0] (line 8, col10) : The Coding provided (http://terminology.cibmtr.org/codesystem/transplant-
center#rc_12002) is not in the value set http://hl7.org/fhir/ValueSet/security-labels, and a code should come from this value set
unless it has no suitable code (note that the validator cannot judge what is suitable).  (error message = Not in value set
http://hl7.org/fhir/ValueSet/security-labels)
  Warning @ Observation.code (line 26, col18) : [Unable to determine whether the provided codes are in the value set
http://hl7.org/fhir/ValueSet/observation-codes because the value set or code system is not known to the validator]
```



```
{
    "resourceType": "Observation",
    "meta": {
        "profile": ["http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables"],
        "security": [
            {
                "system": "http://terminology.cibmtr.org/codesystem/transplant-center",
                "code": "rc_12002"
            }
        ]
    },
    "status": "final",
    "category": [
        {"coding": [
            {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "laboratory"
            }
        ]}
    ],
    "code": {"coding": [
        {
            "system": "http://loinc.org",
            "code": "704-7",
            "display": "Basophils[#/volume] in Blood by Automated count"
        }
    ]},
    "subject": {"reference": "Patient/123"},
    "effectiveDateTime": "2020-07-21T14:17:00Z",
    "valueQuantity": {
        "value": 1,
        "unit": "Thousands Per Microliter",
        "system": "http://unitsofmeasure.org",
        "code": "10*3/uL"
    }
}
```

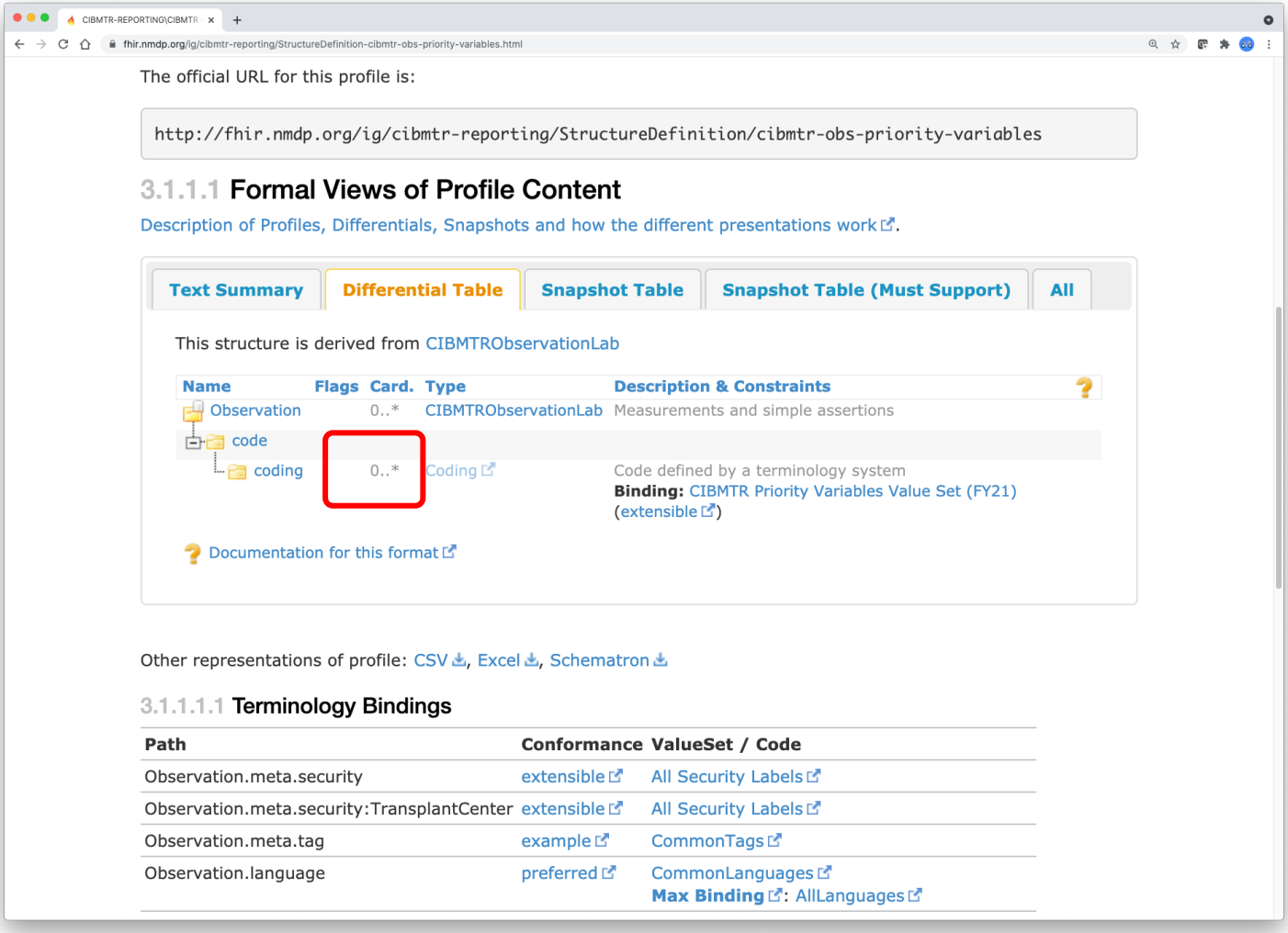

Basophils_profile_wrong-loinc2.json



```
$ java -jar validator_cli.jar Basophils_profile_wrong-loinc2.json -version 4.0.1 -ig ./profiles
  FHIR Validation tool Version 5.4.9 (Git# 296ab05f10f2). Built 2021-07-15T16:29:32.431Z (21 days old)
    Java:   1.8.0_172 from /Library/Java/JavaVirtualMachines/jdk1.8.0_172.jdk/Contents/Home/jre on x86_64 (64bit). 3641MB
  available
    ...
    Params: Basophils_profile_wrong-loinc2.json -version 4.0.1 -ig ./profiles
  Loading
    Load FHIR v4.0 from hl7.fhir.r4.core#4.0.1 - 4575 resources (00:04.0694)
    Load hl7.terminology#2.1.0 - 3767 resources (00:01.0029)
    Terminology server http://tx.fhir.org - Version 1.9.378 (00:01.0648)
    Load ./profiles - 4 resources (00:00.0110)
    Get set...  go (00:00.0036)
  Validating
    Validate Basophils_profile_wrong-loinc2.json 00:01.0179
  Done. Times: Loading: 00:07.0870, validation: 00:01.0179. Memory = 322Mb

  Success: 0 errors, 2 warnings, 1 notes
    Information @ Observation.meta.security[0] (line 8, col10) : Code System URI
  'http://terminology.cibmtr.org/codesystem/transplant-center' is unknown so the code cannot be validated
    Warning @ Observation.meta.security[0] (line 8, col10) : The Coding provided
  (http://terminology.cibmtr.org/codesystem/transplant-center#rc_12002) is not in the value set
  http://hl7.org/fhir/ValueSet/security-labels, and a code should come from this value set unless it has no suitable code
  (note that the validator cannot judge what is suitable).  (error message = Not in value set
  http://hl7.org/fhir/ValueSet/security-labels)
    Warning @ Observation.code.coding[0] (line 27, col10) : The Coding provided (http://loinc.org#704-7) is not in the
  value set http://fhir.nmdp.org/ig/cibmtr-reporting/ValueSet/cibmtr-priority-variables-2021, and a code should come from
  this value set unless it has no suitable code (note that the validator cannot judge what is suitable).  (error message =
  Not in value set http://fhir.nmdp.org/ig/cibmtr-reporting/ValueSet/cibmtr-priority-variables-2021)
```




CIBMTR-REPORTING\CIBMTR
fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition-cibmtr-obs-priority-variables.html
The official URL for this profile is:
http://fhir.nmdp.org/ig/cibmtr-reporting/StructureDefinition/cibmtr-obs-priority-variables
3.1.1.1 Formal Views of Profile Content
Description of Profiles, Differentials, Snapshots and how the different presentations work.
Text Summary
Differential Table
Snapshot Table
Snapshot Table (Must Support)
All
This structure is derived from CIBMTRObservationLab
Name
Flags
Card.
Type
Description & Constraints
Observation
0..*
CIBMTRObservationLab
Measurements and simple assertions
code
coding
0..*
Coding
Code defined by a terminology system
Binding: CIBMTR Priority Variables Value Set (FY21)
(extensible)
Documentation for this format
Other representations of profile: CSV, Excel, Schematron
3.1.1.1.1 Terminology Bindings
Path
Conformance
ValueSet / Code
Observation.meta.security
extensible
All Security Labels
Observation.meta.security:TransplantCenter
extensible
All Security Labels
Observation.meta.tag
example
CommonTags
Observation.language
preferred
CommonLanguages
Max Binding: AllLanguages

Try it out during the hackathon!





Thank you!